

TAMILNADU STATE COUNCIL FOR SCIENCE AND TECHNOLOGY
DOTE Campus, Chennai-600025

STUDENT PROJECTS SCHEME

Proposals invited for 2021-2022

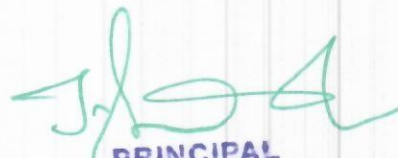
Tamilnadu State Council for Science and Technology has been implementing Student Projects Scheme in order to harness the talent and potential available with students for the benefit of our State. Under this scheme final year UG students (Engineering) / PG Science Students / PG professional Students who have dissertation work in their curriculum may apply. Financial support up to a maximum of Rs.10, 000/- will be provided to each project. Those who want to apply under this scheme are requested to read the instructions carefully and apply only in the prescribed format. The format and instructions in which the proposals are to be submitted is available in the Council's website. (www.tanscst.nic.in).

Two copies of proposals complete in all aspects are to be submitted through proper channel to **The Member Secretary, Tamilnadu State Council for Science and Technology, DOTE Campus, Chennai - 600 025** on or before **24 SEPTEMBER 2021, 5 pm.**

Instructions

1. The scheme is instituted for the purpose of providing assistance in the form of grant to final year students of **UG/PG (Engineering-Civil: Mechanical: :EEE/ECE/EIE/ICE:CSE/IT/Computer Application / Software : Chemical), PG (Professional Courses) and PG (all Sciences)** for scientific research work, with particular relevance to the State of Tamilnadu, for economic and industrial development.
2. Student projects will be eligible for a **maximum grant of Rs.10, 000/-**.
3. **Any One** sector in format (at Sl.NO.4) must be chosen and filled carefully. Incorrect filling or filling more than one sector will lead to rejection of your application.
4. The project team (batch) can have up to a maximum of four students. The project team should send only one proposal.
5. It may be noted that the project proposal should be of original, innovative and novel. They should have utility and applicability for implementation. Kindly note that the evaluation of the project is also based on these factors.
6. The Project Guides / HODs needs to evaluate the project proposals and scrutiny them before sending the project proposals to TNSCST.




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7. The project team **should not** change the topic and should not deviate from the objectives of the sanctioned project proposal. In the event of any such changes, grant will be treated as cancelled and the institution should return the sanctioned amount to TNSCST.
8. Project proposals received will be reviewed and evaluated by a team of experts and those selected will be announced in TNSCST website www.tanscst.nic.in. The project sanction letter and the money will be sent to the Principal/Registrar of the institution with a copy to the Project guide.
9. The sanctioned projects should be completed and the report should be submitted before end of **April 2020**.
10. The Utilization Certificate (UC) and Statement of Expenditure (SE) should be signed by the guide, HOD and Principal/Registrar with official seal as the case may be. The guides are responsible for timely submission of seminar paper, UC and SE.
11. If the guide wishes to leave the institution where the project is based, the Institute/Guide will inform the same to the Council and in consultation with Council, evolve steps to ensure successful completion of the project, before relieving the GUIDE.
12. Project proposals not in the prescribed format and incomplete will not be considered. **Advance copy and soft copy** of the proposals will not be considered.
13. Two copies of the proposal should be signed and forwarded by the Guide through HOD and Principal / Registrar as the case may be to reach the council office on or before **31 AUGUST 2019, 5.00 PM**.
14. Proposals received after the deadline will not be considered and no communication will be entertained in this regard. Postal delays will not be accepted on any ground.

-sd-

MEMBER SECRETARY


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Thiruvallur Dist - 601 206.





**TAMILNADU STATE COUNCIL FOR SCIENCE AND TECHNOLOGY
STUDENT PROJECT PROPOSAL**



1. Name of the Student (s) :

S.No	Name of the Student	E-Mail ID	Phone No.
1	KUMARESAN V	vkumaresan1234@gmail.com	7708811072
2	KARTHICK B	karthick822123@gmail.com	9361880134

2. Name of the Guide

: KAMALKUMAR T

Department / Designation

: Electrical and Electronics Engineering

Institutional Address

: T.J.S Engineering college, T.J.S. Nagar, Peruvoyal, Near
Kavaraipettai, Gummidipoondi Taluk, Thiruvallur-601206

Phone No. & Mobile No.

: 7708811072

3. Project Title

: AI Based Automatic Irrigation System with Android
Application

4. Sector in which your Project
proposal is to be Considered

: Electrical Engineering

5. Project Details

: (write up (max. 3 pages only) should be given for each
item including justification for acquiring / fabricating
equipment's / apparatus in the budget)

i. Introduction

iv. Work Plan

ii. Objectives

v. Budget

iii. Methodology

vi. Any other details

6. Has a similar project been carried
out in your college / elsewhere? If
so, furnish details of the previous
project and highlight the
improvements suggested in the
present one

: NO

CERTIFICATE

This is to certify that Mr. KUMARESAN V, KARTHICK B is a bonafide final year student of U.G. Engineering courses of our college and it is also certified that two copies of utilization certificate and final report along with seminar paper will be sent to the Council after completion of the project by the end of May 2023.

Signature of the Guide

Signature of the HOD

Signature of the Principal/
Head of the Institution

Dr J.PRAKASH, Ph.d., MISTE.,
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N.B : 2 copies of the proposals are to be submitted through proper channel to the Member
Secretary, TNSCST, DOTE Campus, Chennai - 600 025 on or before 07th September 2022,
5 pm.

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INTRODUCTION

In recent years, an explosion in applications of Artificial Intelligence (AI) specifically in the domain of health care, weather forecasting, space programmers, automation, agriculture, etc. has been observed. Across the globe, more than half of the population is involved in agricultural practices. In the arena of agriculture, AI is adopted for the planning of farming activities, sales of agricultural products, weather forecasting, and yield prediction. Farmers are taking advantages of AI and getting higher production than traditional farming techniques. In traditional farming, cultivation was manual process irrigational needs of a crop and its yield was highly dependent on climate factors throughout the globe.

The onset of inventions in industrial revolution and improvement in mechanisms involved in agricultural practices gave relief to miserable life of farmers. But still, it is a labor-intensive practice and dependent on mercy of monsoon. Thus, there is a need of developing a system that automates the agricultural practices such as cultivation and irrigation, predicts yields, and help in better planning for systematizing the demand and supply of agricultural products throughout the globe.

A blend of AI and IoT technologies intelligently gathers relevant information about different crops, their growth rate, and irrigation requirements. These technologies are effective in monitoring frequent climate change, weather forecasting, nutrition deficiency in plants, plants' health monitoring, pest control, weed management, etc.

Researchers have covered a long way in developing subsidiary technology such as geo-positioning satellites (GPS), satellite imagery and sensors for monitoring of yield affecting parameters such as moisture, soil pH level, temperature, etc. Target experimentation at various stages of cultivation such as water resource management, weed elimination, diseases prediction, pest control, estimation of yield production and effective storage of agricultural products is still starving. These challenges can be addressed by applying the techniques of AI and data science.

A voluminous data about agriculture is collected using various sensors such as remote sensors, proximity sensors, temperature/humidity sensors, etc. Advanced Internet facilities disseminate this data to various channels such as the cloud. The introduction of



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online data storage at cloud, advancements in techniques of data processing, and increase in computation power become effective in dealing with challenges identified in agriculture. Use of advanced techniques makes agricultural predictions more precise, reliable, and useful.

OBJECTIVES

The objective is to save the water and supply it to the crop only when it is actually required. The Internet of Things (Iot) with the artificial intelligence technologies are adopted to control the irrigation system using smart phone and cloud base system Smart irrigation technology uses weather data or soil moisture data to determine the irrigation need of the landscape. Smart irrigation technology includes: These products maximize irrigation efficiency by reducing water waste, while maintaining plant health and quality.

Weather Forecasting:

One of the important advantages of AI in agriculture is weather forecasting. With the help of latest AI-based state-of-the-art weather forecasting techniques, it becomes easier for farmers to take appropriate decisions in planning respective crops. Nowadays, many weather forecasting devices are available for predicting the weather.

ABSTRACT

This chapter presents different techniques and applications of Artificial Intelligence for yield prediction and smart irrigation. Timely prediction of irrigation requirements and crop yields is necessary for farmer's welfare and satisfaction. The beforehand prediction significantly contributes to minimizing production cost and maximizing crop yields. The precise prediction of crops' yields is also useful for government, as it is effective in planning various schemes, transport needs, buying mechanisms, storage infrastructure, and liquid position of the economy before actual selling of crop by farmers to market. This chapter acknowledges the past breakthroughs and emerging Artificial Intelligence-based techniques in precision farming specifically for yield prediction and smart irrigation. Artificial Intelligence-based system provides sufficient information about crop yields at



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an early stage and its associated smart irrigation management system is effective in the judicious use of essential resources such as water and energy for agriculture.

METHODOLOGY

S.NO	SOFTWARE	HARDWARE
1.	ARDUINO IDE	ATMega 328p (MP)
2.	PROTEUS 8	Raspberry pi
3	goDaddy WEB Port	Soil moisture

WORK PLAN

Activities:	Work Plan	Status	Starting Date	End Date
Research	Find a Problem statement	To Plan for outcome a problem	28/02/2022	01/02/2022
Evaluation	To Find a solution	To design a solve a problem	02/03/2022	03/03/2022
Selecting Components & IDE	To selected a Hardware & software	Estimated budget	04/03/2022	05/03/2022
Stimulation	Install a software	To design a stimulation part and (RUN a design) it's working.	06/03/2022	07/03/2022
Circuit design	Design a hardware part (circuit diagram)	To order a PCB board (JLPCB)	08/03/2022	10/03/2022
Purchases	To buy a component	All components bought	15/03/2022	18/03/2022
program	To write a program	Run with error	20/03/2022	20/03/2022
Error program	To find out a program	Again, run a program with error	22/03/2022	22/03/2022
Error program	To find out a program	Again, run a program without error	23/03/2022	23/02/2022
Prototype Design	Installing a hardware & software	Install successfully without any interrupt	25/03/2022	28/03/2022
Testing	Testing a without error	Testing successfully	01/04/2022	01/04/2022
Buy & Create a cloud database	Write a cloud program And HTML Program	Import a program to hardware prototype.	02/04/2022	05/04/2022
Final Testing	To testing a prototype in farmer land	To 24 hr monitoring a land, it's working successfully	06/04/2022	07/04/2022



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BUDGET ESTIMATES: SUMMARY**HARDWARE PART**

S.NO	ITEM	QUANTITY	BUDGET In rupee
1.	Arduino Uno	4	3600
2.	Lcd 16*2	1	150
3.	Gsm900A	1	1200
4.	Node MCU (ESP12)	1	600
5.	Soil Moisture Sensor	4	360
6.	Relay 5V(4 channel)	1	200
7.	Digital 3phase Electronic Motor Starter	1	2800
8.	Solar Panel 24v	1	3000
9.	Solar Controller	1	1020
10.	Load Lamp	1	500
11.	PCB Board	3	130
12.	Wire	as required	130
13.	Ultrasonics Sensor	4	240
14.	I ² C Module	1	120
15.	Raspberry pi	1	3000
TOTAL		Rs: 17050	

SOFTWARE PART

		BUDGET
16.	GoDaddy(web hosting) cloud	499 Rs
17.	Arduino IDE	—
Total		Rs: 499

Total(Hardware & Software) = 17050+499 = 17549rs



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11/3/23 தமிழ்நாடு அறிவியல் தொழில்நுட்ப மாதிரி மன்றம்
TAMILNADU STATE COUNCIL FOR SCIENCE AND TECHNOLOGY
(Established by Government of Tamilnadu)
Directorate of Technical Education Campus, Chennai - 600 025.
Ph : 044-22301428, www.tnscst.nic.in

Dr.R. SRINIVASAN, M.Sc., Ph.D., P.I.C.S., M.A.C.S.(USA),
Member Secretary

Lr.No TNSCST/SPS/BS/2022-2023
To
The Principal
T.J.S. Engineering College,
Gummidipoondi- 601 206

03.03.2023



Sir/Madam,

Sub: TNSCST - Student Project Scheme - 2022-2023 - approval intimation-grant release- reg.

With respect to the above scheme, the list of projects approved by the State Council is enclosed along with terms and conditions. You are requested to adhere to terms and conditions such as submission of UC and Seminar Paper on Time.

1	Kamakumar T, Assistant professor, Department of EEE, T.J.S. Engineering College, Gummidipoondi- 601 206	AI based automatic irrigation system with mobile application	Kumaresan V, Karthick B,	EEE-2047	The Principal	Rs 7500/-
						Rs 7500/-

Herewith enclosed the cheque for the approved grant and disburse the grant to the concerned students through the guides at the earliest

Kindly send the utilisation certificate (format enclosed) and seminar paper (Ref.T&C) on completion of the project.

Thanking you,

Yours faithfully,

Member Secretary

Encl: a) Terms & Conditions (T&C)

b) Format of Utilisation Certificate (UC)

c) Cheque for Rs. 7500/- Cheque No. 575087 dt.03.03.2023

Copy to: Individual Guides

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TAMILNADU STATE COUNCIL FOR SCIENCE AND TECHNOLOGY
DOTE campus, Chennai-600 025

STUDENT PROJECT SCHEME 2022-2023

Terms and Conditions for the granted SPS projects

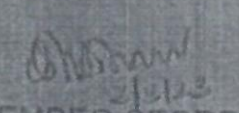
1. Every sanctioned project will be given a Project code Number. Please refer to this number while corresponding with TNSGST.
2. The project team **SHOULD NOT** change the topic of the project and should not deviate from the objectives of the sanctioned proposal. In the event of any such changes, sponsoring will be treated as canceled and the college should return the sanctioned amount to TNSGST.
3. The sanctioned projects should be completed and the reports should be submitted before the end of **MAY 2023**.
4. On completion of the project, a soft copy (CD) of the final project report and **TWO** copies of 2-3 page seminar paper (500 words-MS Word format), utilization certificate (UC) and statement of expenditure (SE) should be sent to The Member Secretary, Tamilnadu State Council For Science and Technology, DOTE Campus, **CHENNAI-600 025**.
5. The seminar paper will be included in the form of **PROCEEDINGS** which will be brought out during the Seminar cum Exhibition, only for those who submit the UC & SE.
6. The Utilization Certificate and Statement of Expenditure should be countersigned by the GUIDE, HOD and Principal/Registrar with an official seal as the case may be.
7. The guides are responsible for the timely submission of **SEMINAR PAPER, UC and SE**.
8. It is mandatory for the project team (**anyone student**) should present and exhibit the findings before the experts in the Seminar cum Exhibition which will be organized during **JULY / AUGUST 2023**.
9. During the Seminar cum Exhibition, "the best project award and certificate" will be presented to the outstanding projects and completion certificates to all.




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10. The council reserves the right to terminate the project at any stage if it is convinced that the grant has not been properly utilized or appropriate progress is not made. In addition, the Council may designate an officer/an Expert to review the work done.
11. If the GUIDE wishes to leave the Institution where the project is based, the Institute/GUIDE will inform the same to Council and in consultation with Council, evolve steps to ensure the successful completion of the project, before relieving the GUIDE.
12. The Council reserves the right to order verification/audit of accounts by any authorized officer. The bills and accounts shall be kept safe for a minimum of 5 years.
13. Unspent money if any should be refunded in the form of DD drawn in favor of The Member Secretary, Tamilnadu State Council for Science and Technology, DOTE Campus, CHENNAI-600 025 payable at CHENNAI.
14. Investigators must acknowledge the Council in reports and technical/scientific papers if published based on the research work done under the project. Investigators are requested to publish some of the research papers emerging out of the project work in peer-reviewed journals Journal's.
15. If the results of the research are to be legally protected by way of patents/copyrights etc. the results should not be published without action being taken to secure legal protection for the research results.
16. The knowledge generated from the project will be the property of TNSCST and should be properly acknowledged. Transfer to technology generated shall be done in consultation with the Council.
17. The recipient organization shall comply, with such other conditions as may be suggested in the 'guidelines' issued in this regard from time to time.
18. The sanctioned grant should not be utilized for the expenses like travel, photocopy, purchase of books, internet charges and report preparation.


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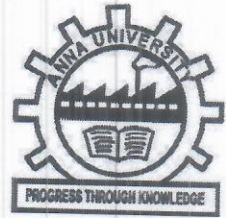

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भारतीय बैंक Indian Bank	Branch : DOTE CAMPUS DOTE OFFICE BUILDINGS GUINDY, CHENNAI IFS Code : IDIB000D060	A/c. Payee Only	VALID FOR THREE MONTHS ONLY दिनांक Date 03/03/2023 D D M M Y Y Y Y
Pay to The Principal, T.J.S. Engineering College, Gummidipoondi			OR BEARER या धारक को
PEES रुपये Seven Thousand Five Hundred only			अदा करें ₹ 7500/-
SB	चल. अ. A/c. No.	479135159	
FOR MEMBER SECRETARY, TAMILNADU STATE COUNCIL SCIENCE & TECHNOLOGY CBS Code: 01636			
00095	PAYABLE AT PAR AT ALL OUR BRANCHES		AUTHORISED SIGNATORY Please sign above
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**AI BASED AUTOMATIC IRRIGATION SYSTEM WITH
ANDROID APPLICATION**

PROJECT REPORT

Submitted by

KUMARESAN V

(112819105003)

KARTHICK B

(112819105304)

In partial fulfilment for the award of the degree

of

BACHELOR OF ENGINEERING

in

ELECTRICAL AND ELECTRONIC ENGINEERING

T.J.S. ENGINEERING COLLEGE, PERUVOYAL

ANNA UNIVERSITY: CHENNAI 600 025

MAY 2023



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Tamil Nadu Dist - Chennai.

ANNA UNIVERSITY: CHENNAI 600 025

BONAFIDE CERTIFICATE

Certified that this project report "AI BASED AUTOMATIC IRRIGATION SYSTEM WITH ANDROID APPLICATION" is the bonafide work of the following students.

KUMARESAN V

(112819105003)

KARTHICK B

(112819105304)

Who carried out the project work under my supervision



SIGNATURE

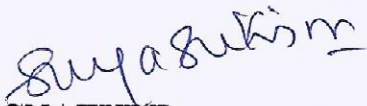
Mr.T.Kamalkumar ,M.E.,

SUPERVISOR

Department of Electrical and Electronics
Engineering

T.J.S. Engineering College,

Peruvoyal.



SIGNATURE

Mrs.M.ShunmugaSankari,M.E.,(Ph.D.),

HEAD OF THE DEPARTMENT

Department of Electrical and Electronics
Engineering

T.J.S. Engineering College,

Peruvoyal.

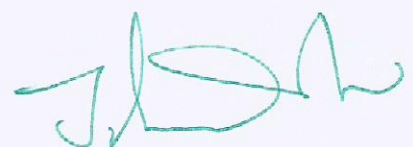
Submitted for viva voce held on 21.06.22 at T.J.S. Engineering College, Peruvoyal.



INTERNAL EXAMINER



EXTERNAL EXAMINER



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ACKNOWLEDGEMENT

"Project is the product out of experience that goes a long way in shaping up a person's calibre. The experience and success one attains is not by oneself but with a group of kind hearts behind."

First and foremost, we express our sincere thanks to honourable Founder and Chairman "KALVI NERI KAVALAR" Shri. T.J.GOVINDARAJAN B.A., Managing Director & Secretary Shri. T.J.ARUMUGAM., Vice Chairman Shri.T.J.DESAMUTHU., Directors Dr.A.PALANI ,B.D.S., Shri. A.VIJAYA KUMAR, M.E., Shri. A.KABILAN B.A. B.L.. M.B.A., Shri. D.DINESH B.Com., L.L.B., Shri. G.TAMILARASAN B.Com., M.B.A., for providing us with adequate infrastructure and congenital academic environment. We also record our sincere thanks to our honourable Principal Dr. J.PRAKASH,M.E.,Ph.D., for his kind support to take up this project.

We express our gratitude to Mrs.M.SHUNMUGA SANKARI.,M.E.,(Ph.D.),Head of the Department of Electrical & Electronics Engineering whose guidance and encouragement has helped us in completing this project work.

We extend our sincere thanks to our guide Mr.T.KAMALKUMAR, M.E., and all other TEACHING FACULTIES and NON-TEACHING STAFF of Department of Electrical & Electronics Engineering for giving the confidence to complete the project successfully by providing the valuable suggestions and interest at every stage of the project.

Further the acknowledgement would be incomplete if we would not mention a word thanks to our most beloved PARENTS and FRIENDS whose continuous support and encouragement all the way through the course has led us to pursue the degree and confidently complete the project.




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ABSTRACT

This chapter presents different techniques and applications of Artificial Intelligence for yield prediction and smart irrigation. Timely prediction of irrigation requirements and crop yields is necessary for farmer's welfare and satisfaction. The beforehand prediction significantly contributes to minimizing production cost and maximizing crop yields. The precise prediction of crops' yields is also useful for government, as it is effective in planning various schemes, transport needs, buying mechanisms, storage infrastructure, and liquid position of the economy before actual selling of crop by farmers to market. This chapter acknowledges the past break through and emerging Artificial Intelligence-based techniques in precision farming specifically for yield prediction and smart irrigation. Artificial Intelligence-based system provides sufficient information about crop yields at an early stage and its associated smart irrigation management system is effective in the judicious use of essential resources such as water and energy for agriculture.



A handwritten signature in green ink, consisting of stylized, overlapping loops and lines.

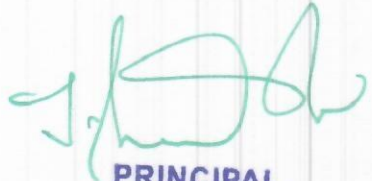
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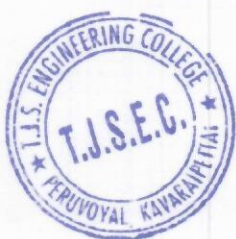
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LIST OF ABBREVIATION

- LCD – Liquid Crystal Display
- DC – Direct Current
- AC – Alternative Current
- IDE – Integrated Development Environment (Software)
- GSM – Global System For Mobile Communication



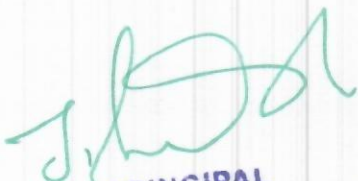

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CHAPTER 1

INTRODUCTION

1.1 GENERAL INTRODUCTION

In recent years, an explosion in applications of Artificial Intelligence (AI) specifically in the domain of health care, weather forecasting, space programmers, automation, agriculture, etc. has been observed. Across the globe, more than half of the population is involved in agricultural practices. In the arena of agriculture, AI is adopted for the planning of farming activities, sales of agricultural products, weather forecasting, and yield prediction. Farmers are taking advantages of AI and getting higher production than traditional farming techniques. In traditional farming, cultivation was manual process irrigational needs of a crop and its yield was highly dependent on climate factors throughout the globe.

The onset of inventions in industrial revolution and improvement in mechanisms involved in agricultural practices gave relief to miserable life of farmers. But still, it is a labour-intensive practice and dependent on mercy of monsoon. Thus, there is a need of developing a system that automates the agricultural practices such as cultivation and irrigation, predicts yields, and help in better planning for systematizing the demand and supply of agricultural products throughout the globe.

A blend of AI and IoT technologies intelligently gathers relevant information about different crops, their growth rate, and irrigation requirements. These technologies are effective in monitoring frequent climate change, weather forecasting, nutrition deficiency in plants, plants' health monitoring, pest control, weed management, etc.

Researchers have covered a long way in developing subsidiary technology such as geo-positioning satellites (GPS), satellite imagery and sensors for monitoring of yield affecting parameters such as moisture, soil pH level, temperature, etc. Target experimentation at various stages of cultivation such as water resource management, weed




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elimination, diseases prediction, pest control, estimation of yield production and effective storage of agricultural products is still starving. These challenges can be addressed by applying the techniques of AI and data science.

A voluminous data about agriculture is collected using various sensors such as remote sensors, proximity sensors, temperature/humidity sensors, etc. Advanced Internet facilities disseminate this data to various channels such as the cloud. The introduction of online data storage at cloud, advancements in techniques of data processing, and increase in computation power become effective in dealing with challenges identified in agriculture. Use of advanced techniques makes agricultural predictions more precise, reliable, and useful.

1.2 AIM OF PROJECT

Water is a very precious resource and must be properly utilized. Agriculture is one of those areas which consume a lot of water. Irrigation is a time-consuming process and must be done on a timely basis.

The aim of the article is to develop an auto irrigation system which measures the moisture of the soil and automatically turns on or off the water supply system.

The project requires very less human involvement once installed. The circuit is based on Raspberry pi controller and also a soil moisture sensor.

A properly configured soil moisture sensor can save up to 60 percent of water used in irrigation. The designed system can be used in turf grass or with small garden plant.

1.3 NEED OF AUTOMATIC IRRIGATION

Automatic irrigation systems are convenient, especially for those who travel. If installed and programmed properly, automatic irrigation systems can even save you money and help in water conservation. Dead lawn grass and plants need to be replaced, and that can be expensive. But the savings from automatic irrigation systems can go beyond that. Watering with a hose or oscillator wastes water. Neither method targets plant roots with any significant degree of precision. Automatic irrigation systems can be programmed to




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discharge more precise amounts of water in a targeted area, which promotes water conservation.

1.4 OBJECTIVES

The objective is to save the water and supply it to the crop only when it is actually required. The Internet of Things (Iot) with the artificial intelligence technologies are adopted to control the irrigation system using smart phone and cloud base system Smart irrigation technology uses weather data or soil moisture data to determine the irrigation need of the landscape. Smart irrigation technology includes: These products maximize irrigation efficiency by reducing water waste, while maintaining plant health and quality.

Weather Forecasting:

One of the important advantages of AI in agriculture is weather forecasting. With the help of latest AI-based state-of-the-art weather forecasting techniques, it become seasier for farmers to take appropriate decisions in planning respective crops. Nowadays, many weather forecasting devices are available for predicting the weather.



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CHAPTER 2

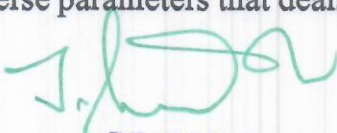
LITERATURE SURVEY

In[1], Automated irrigation control and handle by smart phone is working great. These methods can feed vegetable and made it growth normally. The experimental result shows that the accuracy of sending and receiving command control for irrigation system is 96 percent and accuracy of environment collection is 98 percent. However, measured and collected environment data from sensor to manage vegetable crop including water level, air temperature and air humidity were not enough for making the intelligence decision. Watering volume is a little faulty which is sometimes caused by overwatering. The error related to many factors such as rain, soil type, and specification of vegetable crops.

In[2], The Wireless Sensors Network (WSN) is nowadays widely used to build decision support systems to overcome many problems in the real-world. One of the most interesting fields having an increasing need of decision support systems is precision agriculture (PA). This approach provides real-time information about the lands and crops that will help farmers make right decisions. Using the basic principles of Internet and WSN technology, precision agriculture systems based on the internet of things (IOT) technology is explained in detail especially on the hardware architecture, network architecture and software process control of the precision irrigation system.

In[3], The internet of things (IoT) is an emerging technology that portrays the possibility of physical objects being connected with the internet and having the capacity to distinguish themselves from different gadgets. One of the significant uses of IoT is smart agriculture. Real task of smart agriculture is to cultivate & convey it to the end customers with the most ideal cost and quality. The quality of the crop essentially depends upon the factors like soil moisture, adequate temperature & the presence of desired chemicals within the soil. This paper fundamentally manages the convergence of agriculture with the IoT by utilising different sensors with the end goal to check the diverse parameters that deals with



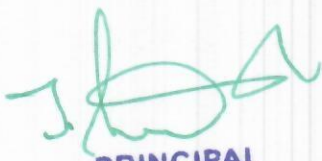

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viable production of crops, prompting automation of water supply to the fields and to check the continuous environmental temperature, humidity and moisture substance of the soil which additionally encourages us to productively use electric power

In[4], Agriculture sector being the backbone of the Indian economy deserves security. Security not in terms of resources only but also agricultural products needs security and protection at very initial stage, like protection from attacks of rodents or insects, in fields or grain stores. Such challenges should also be taken into consideration. Security systems which are being used now a days are not smart enough to provide real time notification after sensing the problem. The integration of traditional methodology with latest technologies as Internet of Things and Wireless Sensor Networks can lead to agricultural modernization.

In[5], Smart Agriculture helps to reduce wastage, effective usage of fertilizer and thereby increase the crop yield. In this work, a system is developed to monitor crop-field using sensors (soil moisture, temperature, humidity, Light) and automate the irrigation system. The data from sensors are sent to Web server database using wireless transmission. In server database the data are encoded in JSON format. The irrigation is automated if the moisture and temperature of the field falls below the brink. In greenhouses light intensity control can also be automated in addition to irrigation.




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CHAPTER 3

EXISTING AND PROPOSED SYSTEM

3.1 EXISTING SYSTEM

An automatic irrigation system does the work quite efficiently and with a positive impact on the place where it is installed. Once it is installed in the agricultural field, the water distribution to crops and nurseries becomes easy and doesn't require any human support to perform the operations permanently. Sometimes automatic irrigation can also be performed by using mechanical appliances such as clay pots or bottle irrigation system. It's very hard to implement irrigation systems because they are very expensive and complex in their design. By taking some basic points into considerations from experts' support, we have implemented some projects on automatic irrigation system by using different technologies.

In this project, we are describing about three types of irrigation systems that work automatically and each system is an advancement of the previous one as we go from first system to the next, and so on.

3.1.1 Disadvantages of Existing System

Limitations of Existing System:

- Overflow of water can cause:
overflow of water, if stagnated, may lead to diseases causing organisms likes providing breeding places for mosquitos and other organisms.
- Over –irrigation leads to water loss.
- Results in colder and damper climate causing outbreak of diseases like malaria.
- Water logging.



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3.2 PROPOSED SYSTEM

Whenever we go out of town for few days, we always used to worry about our plants as they need water on regular basis. So here we are making Automatic Plant Irrigation System using GSM and Raspberry pi technology, which automatically provides water to your plants and keep you updated by sending message to your cell phone.

- In This Plant Watering System, data collection and processing are being done in the background using MATLAB analytics techniques. Water pump gets automatically off when system finds enough moisture in the soil. Two types mode of operation:

1. Automatic mode
2. Manual operation mode

- Radar Monitor:

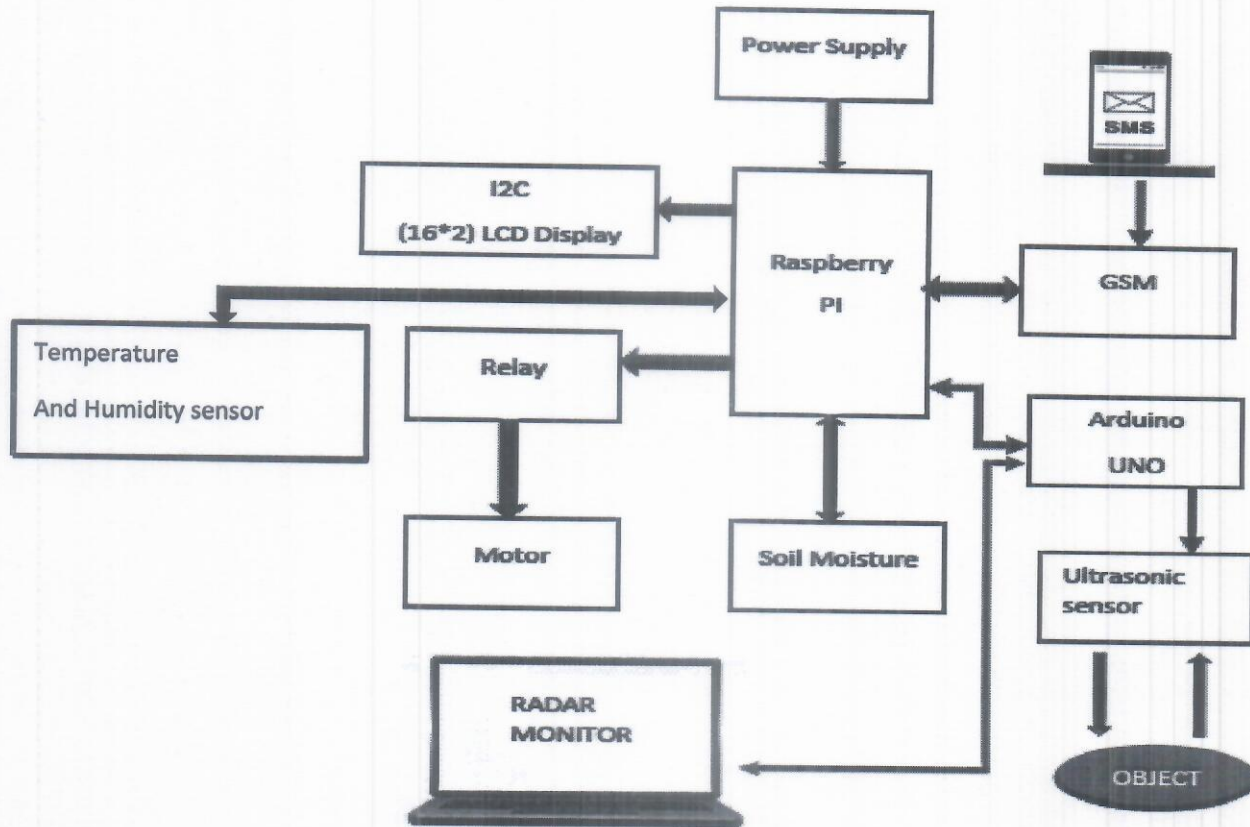
To prevent animals from ruining the crops by navigating the animals.

Whenever system switched On or off the pump, a message is sent to the user via GSM module, updating the status of water pump and soil moisture. This system is very useful in Farms, gardens, home etc. This system is completely automated and there is no need for any human intervention.



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3.2.1 Block Diagram of Proposed System



FigNo: 3.1 Block Diagram of Proposed System

Description of The Proposed System:

Step 1: "CONDITION" The Message from user mobile to gsm module.

Step 2: The analysis data from humidity sensor and soil moisture sensor from GSM Module to user mobile.

Step 3: "AREA 1" message form user to GSM Area 1 ONN Area 2 OFF.

Step 4: else(if wants AREA 2 ONN) "AREA 2" Message from user to GSM Area 1 ON Area 2 OFF.



Step 5: "RUN TIME 15 MIN" message from user to gsm .

Step 6: "MOTOR RUN TIME 15 MIN" Message from GSM to user (motor running time).

Step 7: "MOTOR OFF" Message from user to GSM (motor off).

3.2.2 Advantages of Proposed System

- It can be an automated sprinkler system.
- Irrigation systems will increase the value of your home.
- You can help prevent weeds and disease in your plants.
- Maintain soil nutrient balance.
- To prevent wild animals from destroying crops.
- You are in complete control.



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Features:

Hardware:

- Quad core 64-bit ARM-Cortex A72 running at 1.5GHz
- 1, 2 and 4 Gigabyte LPDDR4 RAM options
- H.265 (HEVC) hardware decode (up to 4Kp60)
- H.264 hardware decode (up to 1080p60)
- VideoCore VI 3D Graphics
- Supports dual HDMI display output up to 4Kp60

Interfaces:

- 802.11 b/g/n/ac Wireless LAN
- Bluetooth 5.0 with BLE
- 1x SD Card
- 2x micro-HDMI ports supporting dual displays up to 4Kp60 resolution
- 2x USB2 ports
- 2x USB3 ports
- 1x Gigabit Ethernet port (supports PoE with add-on PoE HAT)
- 1x Raspberry Pi camera port (2-lane MIPI CSI)
- 1x Raspberry Pi display port (2-lane MIPI DSI)
- 28x user GPIO supporting various interface options:



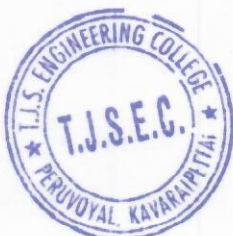
- Up to 6x UART
- Up to 6x I2C
- Up to 5x SPI
- 1x SDIO interface
- 1x DPI (Parallel RGB Display)
- 1x PCM
- Up to 2x PWM channels
- Up to 3x GPCLK outputs

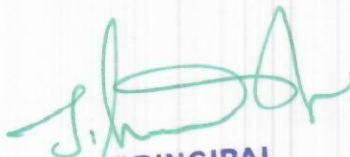
Software:

- ARMv8 Instruction Set
- Mature Linux software stack
- Actively developed and maintained
 - Recent Linux kernel support
 - Many drivers up streamed
 - Stable and well supported userland
 - Availability of GPU functions using standard APIs

Electrical Specification:

Caution! Stresses above those listed in Table 2 may cause permanent damage to the device. This is a stress rating only; functional operation of the device under these or any other conditions above those listed in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.




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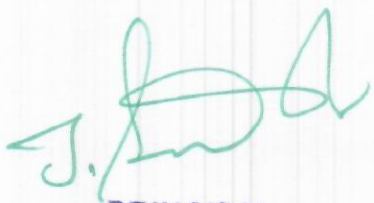
Power Requirements:

The Pi4B requires a good quality USB-C power supply capable of delivering 5V at 3A. If attached downstream USB devices consume less than 500mA, a 5V, 2.5A supply may be used.

Peripherals:

- GPIO Interface
- Display Parallel Interface (DPI)
- SD/SDIO Interface
- Camera and Display Interfaces
- USB
- HDMI
- Audio and Composite (TV Out)
- Temperature Range and Thermals




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PROGRAMMING ENVIRONMENT:

Burn firmware

Enter the official website and follow the instructions to download and burn the firmware:

<https://www.raspberrypi.com/documentation/microcontrollers/micropython.html>

Operation instructions:

Press and hold the BOOTSEL button, insert the USB cable connected to the into the USB port of the computer, a new U disk folder will pop up on the computer, drag and drop the “UF2” file just downloaded to the folder, the Raspberry Pi will restart automatically. In this way, the firmware burning is completed.

Install programming software

Enter the software official website “<https://thonny.org/>” to download the software, it is best to download the latest version, otherwise the Raspberry Pi may not be supported;

Operation instructions:

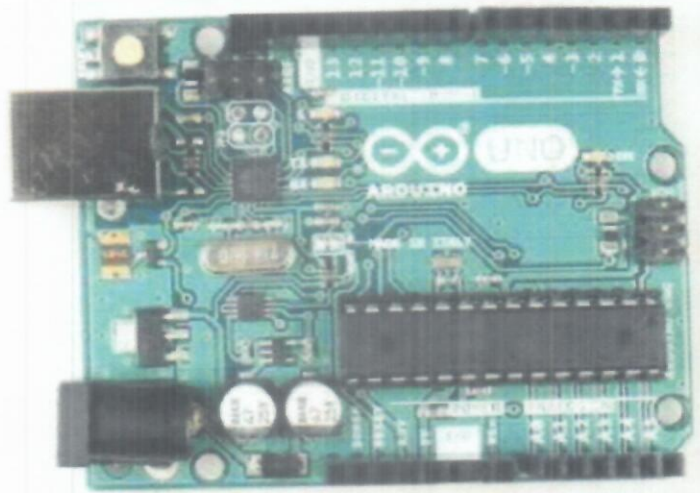
Install Thonny, open Thonny software after installation, open “Tools -> Settings -> Interpreter”, select MicroPython (Raspberry Pi) interpreter, and select the serial port number of Raspberry Pi at the serial port (If the board has been connected to the computer, the software will generally automatically detect the serial port number) and click OK, you can see that the files in the Raspberry Pi are displayed at the bottom left of the software; if the file tree on the left is not displayed, you can check the “view -> file”.




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4.2 ARUDINO UNO R3

Arduino Uno R3 is one kind of ATmega328P based microcontroller board. It includes the whole thing required to hold up the microcontroller; just attach it to a PC with the help of a USB cable, and give the supply using AC-DC adapter or a battery to get started. The term Uno means “one” in the language of “Italian” and was selected for marking the release of Arduino’s IDE 1.0 software. The R3 Arduino Uno is the 3rd as well as most recent modification of the Arduino Uno. Arduino board and IDE software are the reference versions of Arduino and currently progressed to new releases. The Uno-board is the primary in a sequence of USB-Arduino boards,& the reference model designed for the Arduino platform.



Arduino Uno R3

FigNo:4.2 Arduino UNO R3

Arduino Uno R3 Specifications

The Arduino Uno R3 board includes the following specifications.

- It is an ATmega328P based Microcontroller
- The Operating Voltage of the Arduino is 5V
- The recommended input voltage ranges from 7V to 12V
- The i/p voltage (limit) is 6V to 20V




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- Digital input and output pins-14
- Digital input & output pins (PWM)-6
- Analog i/p pins are 6
- DC Current for each I/O Pin is 20 mA
- DC Current used for 3.3V Pin is 50 mA
- Flash Memory -32 KB, and 0.5 KB memory is used by the boot loader
- SRAM is 2 KB
- EEPROM is 1 KB 4
- The speed of the CLK is 16 MHz
- In Built LED
- Length and width of the Arduino are 68.6 mm X 53.4 mm
- The weight of the Arduino board is 25 g

Arduino Uno R3 Pin Diagram

The Arduino Uno R3 pin diagram is shown below. It comprises 14-digit I/O pins. From these pins, 6-pins can be utilized like PWM outputs. This board includes 14 digital input/output pins, Analog inputs-6, a USB connection, quartz crystal-16 MHz, a power jack, a USB connection, resonator-16Mhz, a power jack, an ICSP header an RST button.



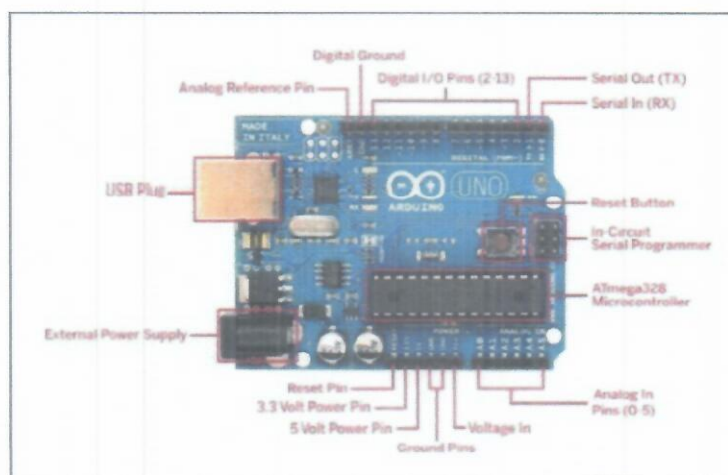


Fig No:4.2.1 Arduino Uno R3 Pin Diagram

Power Supply

The power supply of the Arduino can be done with the help of an exterior power supply otherwise USB connection. The exterior power supply (6 to 20 volts) mainly includes a battery or an AC to DC adapter. The connection of an adapter can be done by plugging a center-positive plug (2.1mm) into the power jack on the board. The battery terminals can be placed in the pins of Vin as well as GND. The power pins of an Arduino board include the following.

Vin: The input voltage or Vin to the Arduino while it is using an exterior power supply opposite to volts from the connection of USB or else RPS (regulated power supply). By using this pin, one can supply the voltage.

5Volts: The RPS can be used to give the power supply to the microcontroller as well as components which are used on the Arduino board. This can approach from the input voltage through a regulator.

3V3: A 3.3 supply voltage can be generated with the onboard regulator, and the highest draw current will be 50 mA.



GND:GND (ground) pins

Memory

The memory of an ATmega328 microcontroller includes 32 KB and 0.5 KB memory is utilized for the Boot loader), and also it includes SRAM-2 KB as well as EEPROM-1KB.

Input and Output

We know that an Arduino Uno R3 includes 14-digital pins which can be used as an input otherwise output by using the functions like pin Mode (), digital Read(), and digital Write(). These pins can operate with 5V, and every digital pin can give or receive 20mA, & includes a 20k to 50k ohm pull up resistor. The maximum current on any pin is 40mA which cannot surpass for avoiding the microcontroller from the damage. Additionally, some of the pins of an Arduino include specific functions.

Serial Pins

The serial pins of an Arduino board are TX (1) and RX (0) pins and these pins can be used to transfer the TTL serial data. The connection of these pins can be done with the equivalent pins of the ATmega8 U2 USB to TTL chip.

External Interrupt Pins

The external interrupt pins of the board are 2 & 3, and these pins can be arranged to activate an interrupt on a rising otherwise falling edge, a low-value otherwise a modify in value

PWM Pins

The PWM pins of an Arduino are 3, 5, 6, 9, 10, & 11, and gives an output of an 8-bit PWM with the function analog Write ().



SPI (Serial Peripheral Interface) Pins

The SPI pins are 10, 11, 12, 13 namely SS, MOSI, MISO, SCK, and these will maintain the SPI communication with the help of the SPI library.

LED Pin

An Arduino board is inbuilt with a LED using digital pin-13. Whenever the digital pin is high, the LED will glow otherwise it will not glow.

TWI (2-Wire Interface) Pins

The TWI pins are SDA or A4, & SCL or A5, which can support the communication of TWI with the help of Wire library.

AREF (Analog Reference) Pin

An analog reference pin is the reference voltage to the inputs of an analog i/p using the function like analog Reference().

Reset (RST) Pin

This pin brings a low line for resetting the microcontroller, and it is very useful for using an RST button toward shields which can block the one over the Arduino R3 board.

Communication

The communication protocols of an Arduino Uno include SPI, I2C, and UART serial communication.

UART

An Arduino Uno uses the two functions like the transmitter digital pin1 and the receiver digital pin0. These pins are mainly used in UART TTL serial communication.



I2C

An Arduino UNO board employs SDA pin otherwise A4 pin & A5 pin otherwise SCL pin is used for I2C communication with wire library. In this, both the SCL and SDA are CLK signal and data signal.

SPI Pins

The SPI communication includes MOSI, MISO, and SCK.

MOSI (Pin11)

This is the master out slave in the pin, used to transmit the data to the devices

MISO (Pin12)

This pin is a serial CLK, and the CLK pulse will synchronize the transmission of which is produced by the master.

SCK (Pin13)

The CLK pulse synchronizes data transmission that is generated by the master. Equivalent pins with the SPI library is employed for the communication of SPI. ICSP (in-circuit serial programming) headers can be utilized for programming ATmega microcontroller directly with the boot loader.

Arduino Uno R3 Programming

- The programming of an Arduino Uno R3 can be done using IDE software. The microcontroller on the board will come with pre-burned by a boot loader that permits to upload fresh code without using an exterior hardware programmer.
- The communication of this can be done using a protocol like STK500.
- We can also upload the program in the microcontroller by avoiding the boot loader using the header like the In-Circuit Serial Programming.



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4.3 SOFTWARE APPLICATION:

Program for raspberry pi:

To build interesting projects with Raspberry Pi, you need to learn how to program it first. Raspberry Pi supports two types of programming languages: C++ and MicroPython. C++ is a general-purpose language developed on the basis of the C language. It is often used in the development of ESP32 and Arduino, so most microcontroller users are familiar with this language.



MicroPython

Python is an interpreted general-purpose language. Since the first version of Python was designed and released by Guido van Rossum of the National Research Institute for Mathematics and Computer Science in 1991, Python has rapidly become one of the most popular languages in the world with its concise syntax, excellent code readability and hot community resources.



Fig No:4.3 MicroPython

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However, the powerful Python is not omnipotent. Although Python performs well on desktop computers, large terminals and servers, it cannot be deployed on microcontrollers with limited resources and small memory. As a result, MicroPython, a Python-based interpreted language specially designed for microcontrollers, was born. MicroPython optimizes Python cleverly, and the optimized MicroPython can run perfectly in all kinds of resource-constrained microcontrollers. MicroPython also inherits various advantages of Python, such as its ease of use.

Thonny:

Thonny is a Python/ MicroPython IDE (integrated development environment) for programming beginners, developed by the University of Tartu in Estonia.

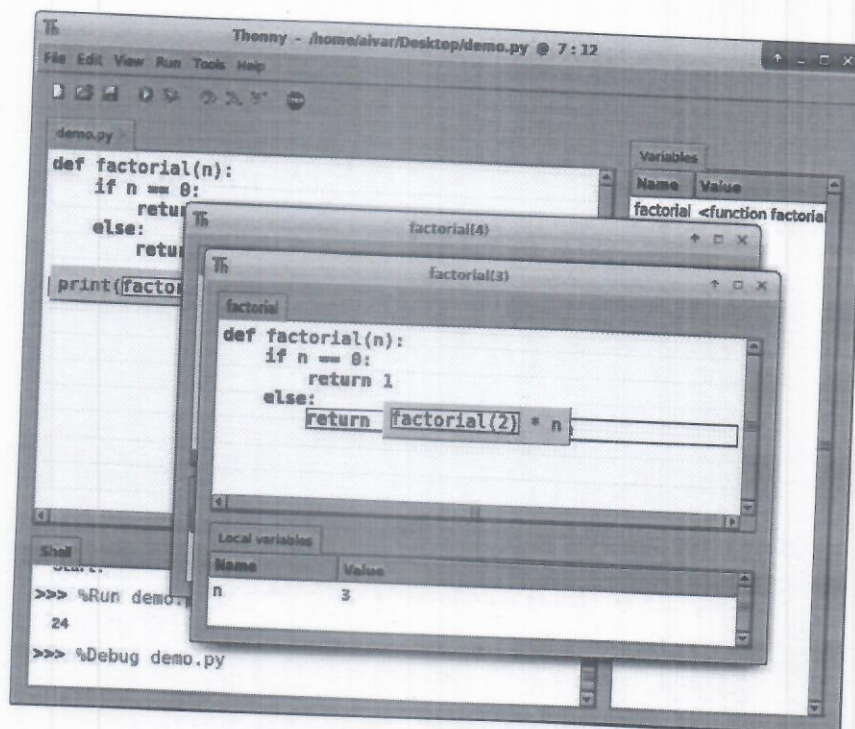


Fig No:4.3.1 MicroPython IDE



The platform integrates many tools that users need to use when developing. Its interface is simple and easy to understand, which is very suitable for beginners.

Program for Arduino uno 2.0:

The Arduino IDE 2.0 is an improvement of the classic IDE, with increased performance, improved user interface and many new features, such as autocompletion, a built-in debugger and syncing sketches with Arduino Cloud.

Overview:

The Arduino IDE 2.0 features a new sidebar, making the most commonly used tools more accessible.

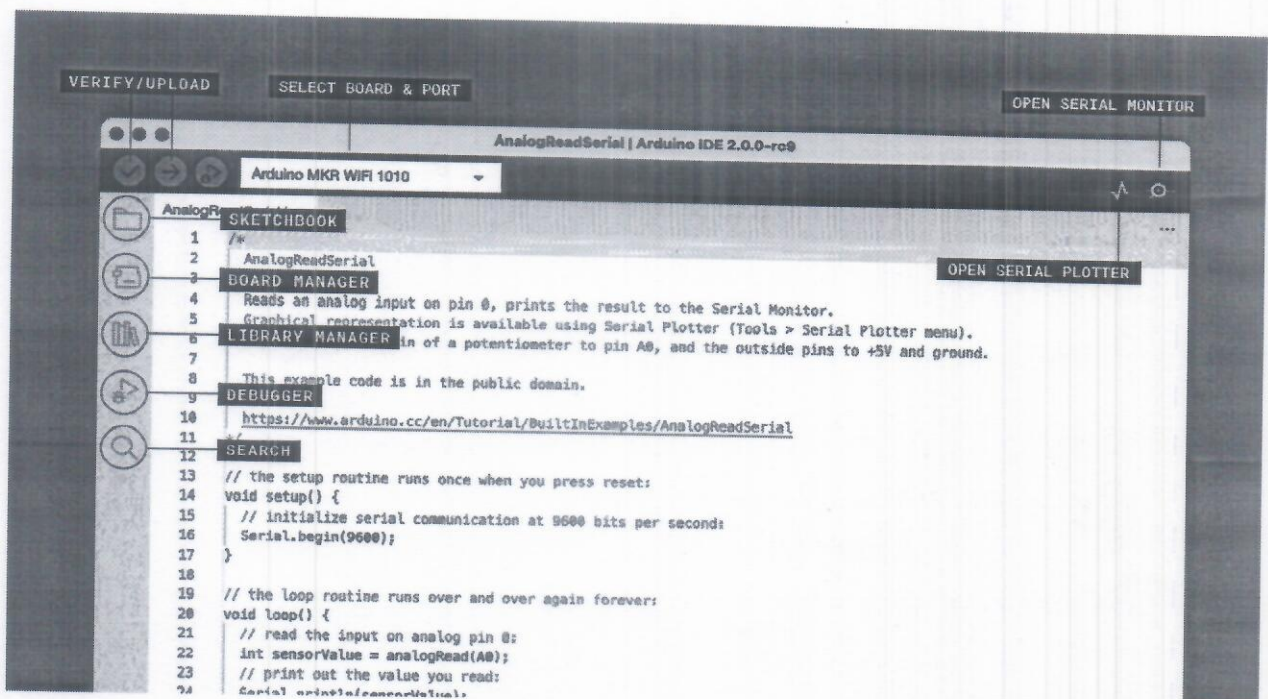


Fig No:4.3.2 Arduino IDE

- **Verify / Upload** - compile and upload your code to your Arduino Board.
- **Select Board & Port** - detected Arduino boards automatically show up here, along with the port number.
- **Sketchbook** - here you will find all of your sketches locally stored on your computer. Additionally, you can sync with the Arduino Cloud, and also obtain your sketches from the online environment.
- **Boards Manager** - browse through Arduino & third-party packages that can be installed. For example, using a MKR Wi-Fi 1010 board requires the package installed.
- **Library Manager** - browse through thousands of Arduino libraries, made by Arduino & its community.
- **Debugger** - test and debug programs in real time.
- **Search** - search for keywords in your code.
- **Open Serial Monitor** - opens the Serial Monitor tool, as a new tab in the console.

IT is an open-source computer hardware and software company, project and user community that designs and manufactures microcontroller-based kits for building digital devices and interactive objects that can sense and control the physical world. The Arduino integrated development environment (IDE) is a cross-platform application written in Java, and is derived from the IDE for the Processing programming language and the Wiring projects. which also supports the languages C and C++. A typical Arduino C/C++ sketch consist of two functions that are compiled and linked with a program stub `main ()` into an executable cyclic executive program:

- **Setup ()**: a function that runs once at the start of a program and that can initialize settings.
- **Loop ()**: a function called repeatedly until the board powers off.




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After compiling and linking with the GNU tool chain, also included with the IDE distribution, the Arduino IDE employs the program avrdude to convert the executable code into a text file in hexadecimal coding that is loaded into the Arduino board by a loader program in the board's firmware. Arduino more straight forward by allowing the use of an ordinary computer as the programmer. Currently, optiboot loader is the default boot loader installed on Arduino UNO. At a conceptual level, when using the Arduino integrated development environment, all boards are programmed over a serial connection. Its implementation varies with the hardware version. Some serial Arduino boards contain a level shifter circuit to convert between RS-232 logic levels and transistor-transistor logic (TTL) level signals. Current Arduino boards are programmed via Universal Serial Bus (USB), implemented using USB-to-serial adapter chips such as the FTDI FT232.



HARDWARE PROTOTYPE:

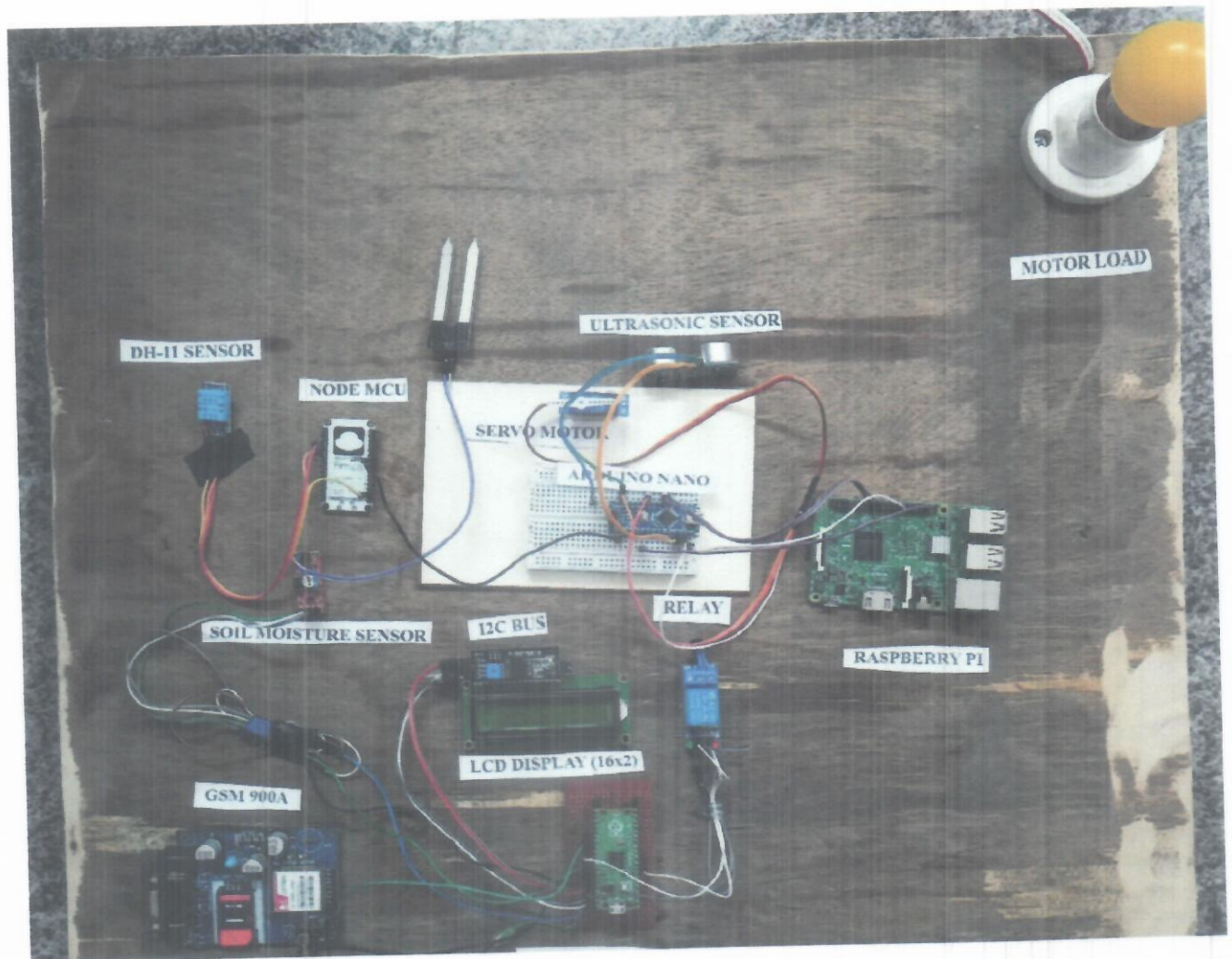


Fig No: 4.4 Hardware Prototype

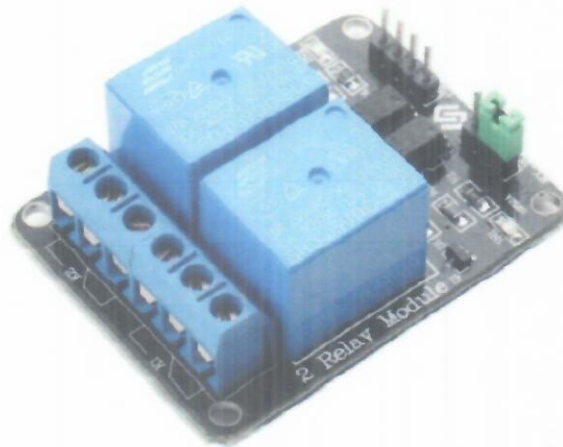


CHAPTER 5

HARDWARE IMPLEMENTATION

5.1 RELAY BOARD

A relay is an electrically operated switch. Several relays use a magnet to automatically operate a switch, however alternative in operation principles are used, like solid state relays. Relays are used wherever it's necessary to regulate a circuit by a separate low-power signal, or wherever many circuits should be controlled by one signal. The essential relays were handling in long distance communicate circuits as amplifiers, they unbroken the signal coming back in from one circuit and re-transmitted it on another circuit.



FigNo:5.1 RELAY BOARD



5.2 SUBMERSIBLE PUMP

Electric submersible pumps are multistage centrifugal pumps operating in a vertical position. Liquids, accelerated by the impeller, lose their kinetic energy in the diffuser, where a conversion of kinetic to pressure energy takes place. This is the main operational mechanism of radial and mixed flow pumps. In the HSP, the motor is a hydraulic motor rather than an electrical motor, and may be closed cycle (keeping the power fluid separate from the produced fluid) or open cycle (mingling the power fluid with the produced fluid downhole, with surface separation).

The pump shaft is connected to the gas separator or the protector by a mechanical coupling at the bottom of the pump. Fluids enter the pump through an intake screen and are lifted by the pump stages. Other parts include the radial bearings (bushings) distributed along the length of the shaft, providing radial support to the pump shaft. An optional thrust bearing takes up part of the axial forces arising in the pump, but most of those forces are absorbed by the protector's thrust bearing.

There are also screw-type submersible pumps, there is a steel screw which is used as a working element in them. The screw allows the pump to work in water with a high sand content and other mechanical

APPLICATION

Submersible pumps are found in many applications. Single stage pumps are used for drainage, sewage pumping, general industrial pumping and slurry pumping. They are also popular with Pond filters. Multiple stage submersible pumps are typically lowered down a borehole, and most typically used for residential, commercial, municipal and industrial water extraction (abstraction), water well sand in oil wells.



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Other uses for submersible pumps include treatment plants, seawater handling, fire fighting(since it is retardant cable), water well and deep well drilling, offshore rigs, artificial mine dewatering, and irrigation systems.

Pumps in electrical hazardous locations used for combustible liquids or for water that may be contaminated with combustible liquids must be designed not to ignite the liquid or vapors.

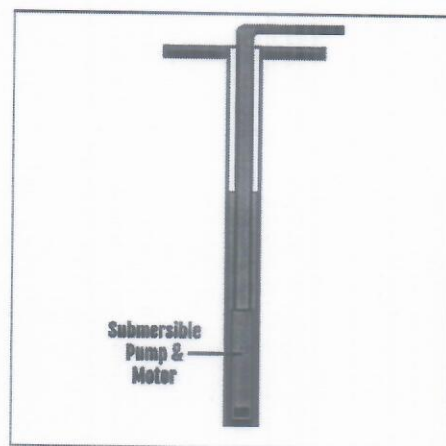


Fig.No. 5.2 EXAMPLE DIAGRAM FOR SUBMERSIBLE PUMP

5.3 LCD DISPLAY

- The LCDs have a parallel interface, meaning that the microcontroller has to manipulate several interface pins at once to control the display. The interface consists of the following pins:
- A **register select (RS)** pin that controls where in the LCD's memory you're writing data to. You can select either the data register, which holds what goes on the screen, or an instruction register, which is where the LCD's controller looks for instructions on what to do next.
 - A Read/Write (R/W) pin that selects reading mode or writing mode
 - An Enable pin that enables writing to the registers



- 8 data pins (D0 -D7). The states of these pins (high or low) are the bits that you're writing to a register when you write, or the values you're reading when you read.
- There's also a display contrast pin (Vo), power supply pins (+5V and Gnd) and LED Backlight (Bkl+ and Bkl-) pins that you can use to power the LCD, control the display contrast, and turn on and off the LED backlight, respectively.
- The process of controlling the display involves putting the data that form the image of what you want to display into the data registers, then putting instructions in the instruction register. The LiquidCrystal Library simplifies this for you so you don't need to know the low-level instructions.
- The Hitachi-compatible LCDs can be controlled in two modes: 4-bit or 8-bit. The 4bit mode requires seven I/O pins from the Arduino, while the 8-bit mode requires 11 pins. For displaying text on the screen, you can do most everything in 4-bit mode, so example shows how to control a 16x2 LCD in 4-bit mode.

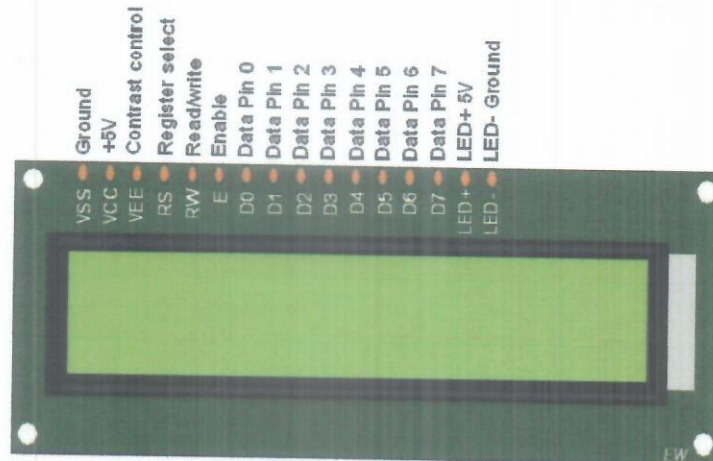


Fig.No5.3 LCD Display



5.4 SOIL MOISTURE SENSOR

The Soil Moisture Sensor is used to measure the volumetric water content of soil. This makes it ideal for performing experiments in courses such as soil science, agricultural science, environmental science, horticulture, botany, and biology. Use the Soil Moisture Sensor to:

- Measure the loss of moisture over time due to evaporation and plant uptake.
- Evaluate optimum soil moisture contents for various species of plants.
- Monitor soil moisture content to control irrigation in greenhouses.
- Enhance your Bottle Biology™ experiments.

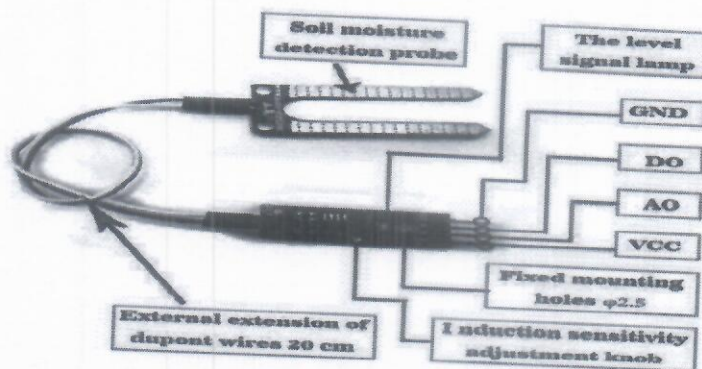


Fig No 5.4 Soil Moisture Sensor

Principles of Water Level sensor:

A water level sensor or liquid level detector is a very useful device for measuring or detecting a required level of water, liquid or fluid in a tank, pool, well, aquarium, washing machines etc. The water level indicator shown here is very simple but can perform a great task as a monitor for desired level of any liquid, it will activate the LED when the



two probes shown in the circuit will detect liquid. The circuit is using only four components and can be built in few minutes if all parts are available with you.

Plant Soil Moisture Tester

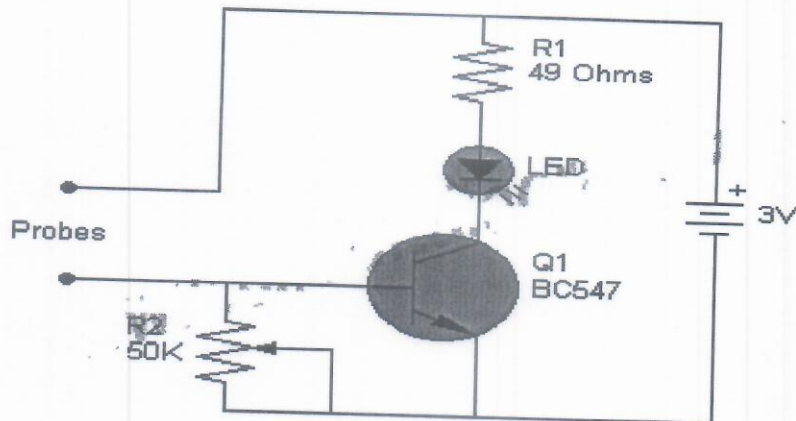


Fig.No 5.4.1 Soil Moisture Circuit

Working of the circuit is simple, the circuit is based on a NPN transistor BC547 which is working as a switch, when water level will touch the two probes the transistor become switched on and the LED in the circuit will become activated. The circuit can be operated with two 1.5V batteries of any size.



5.5 GSM :

SIM900A GSM Module is the smallest and cheapest module for GPRS/GSM communication. It is common with Arduino and microcontroller in most of embedded application. The module offers GPRS/GSM technology for communication with the uses of a mobile sim. It uses a 900 and 1800MHz frequency band and allows users to receive/send mobile calls and SMS. The keypad and display interface allows the developers to make the customize application with it. Furthermore, it also has modes, command mode and data mode. In every country the GPRS/GSM and different protocols/frequencies to operate. Command mode helps the developers to change the default setting according to their requirements.



Fig No: 5.5 GSM

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SIM900A Pin Configuration:

The Module SIM900A looks like a single chip but it has a bunch of features that can help to build almost many commercial applications. Although, there are a total of 68 pins on SIM900A and using these pins helps to build the applications. But we will need few pins if you use a module for interfacing with Arduino. We lists details of pinout diagram in next section.

CONNECTION WITH CONTROLLER

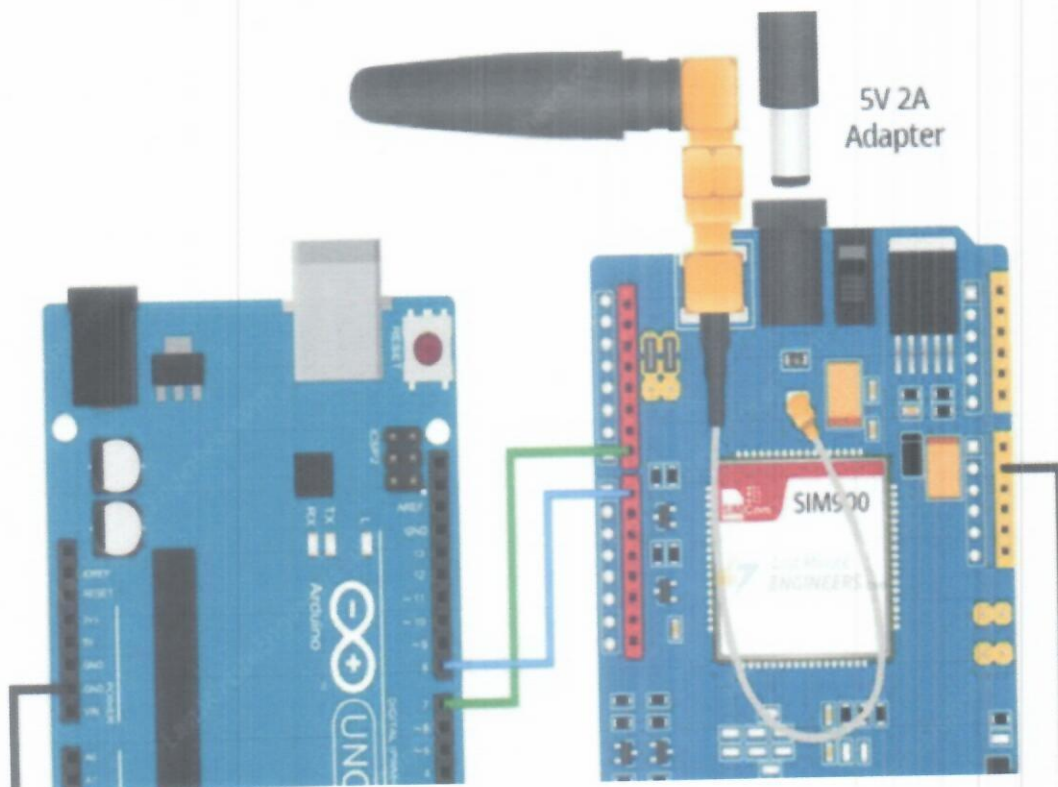
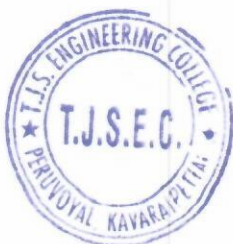


Fig No: 5.5.1 GSM Connection



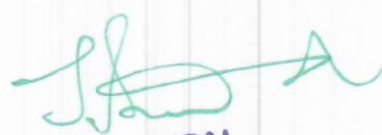
Interfacing GSM With MC:

- The connection is simple. we shall use the hardware serial port of MC (pin 7 & pin 6).
- Pin 7 (RX) OF MC is connected to RX pin of GSM
- Pin 6 (TX) of MC is connected to TX pin of GSM
- General rule is always TX to RX & RX to TX.

Do not power the GSM from MC. use a separate adapter of minimum 1 Amp rating

- Make GND pins of both GSM & MC common.
- To start with place a valid SIM to the holder on the GSM board. connect the power adapter to GSM.
- Now dial a call from another phone to the SIM number & ensure that you get a ring back tone.
- This is the initial test to confirm that your GSM is ready to accept commands from MC.




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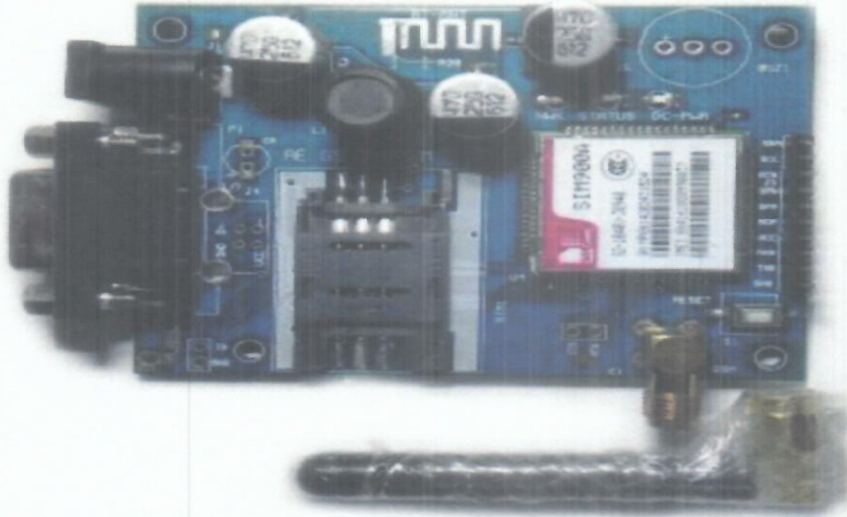


Fig No : 5.5.3 GSM 900A

SIM900A GSM Module Pin Configuration Description GPIO Pins:

The GPIO pins help to perform the simple and advance I/O function. All pins give the maximum output equal to the power supply which is useable to control most of the devices like sensors and other modules. All GPIO pins in SIM900A are:

- GPIO1 – Pin40
- GPIO2 – Pin41
- GPIO3 – Pin42
- GPIO4 – Pin43
- GPIO5 – Pin44
- GPIO6 – Pin47
- GPIO7 – Pin48
- GPIO8 – Pin49
- GPIO9 – Pin50
- GPIO10 – Pin51
- GPIO11 – Pin67
- GPIO12 – Pin68



SIM900A GSM Module Block Diagram:

The following diagram is describing the SIM900A internal structure of the module.

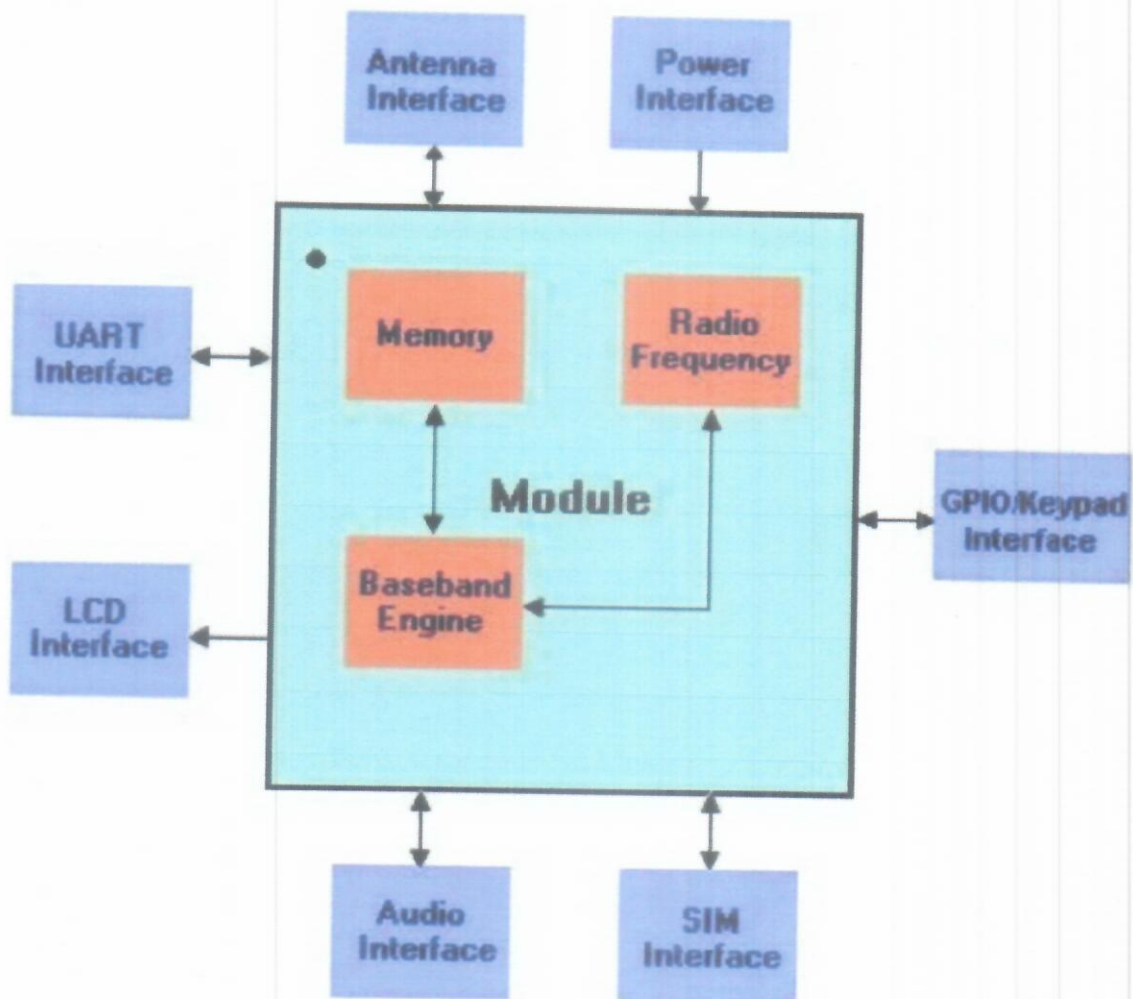


Fig No : 5.5.4 SIM900A GSM Module Block Diagram



Operation:

The operation done using GSM are

- Test the simple AT command.
- Find out the International mobile station equipment identify (IMEI) number of the GSM modem.
- Connect a call to a GSM mobile number (i.e) dial a number.
- Send a text to that mobile number.
- These operations are done using AT commands. The provision of these four operations has been provided by means of four tactile switches. Each switch corresponds to each of the above functions.
- AT+CMGF message format: this command is used to select SMS protocol data unit (PDU) mode or SMS text mode.
- At+CMGS send message: sends a message from GSM to the network (SMS+CMGS).The message reference value is returned to the GSM on successful message delivery.
- Sending can be cancelled with the character must be used to indicate the ending of the message body.



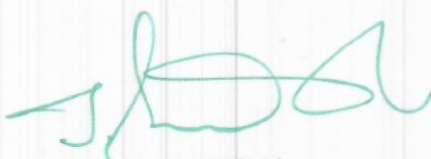

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SIM900A GSM Module Main Features:

FEATURES	DETAIL
Power Input	3.4V to 4.5V
Operating Frequency	EGSM900 and DCS1800
Transmitting Power Range	2V for EGSM900 and 1W for DCS1800
Data Transfer Link	Download: 85.6kbps, Upload:42.8kbps
SMS	MT, MO, CB, Text and PDU mode.
Antenna Support	Available
Audio Input/output	Available
Serial Port	I2C and UART
Serial Debug Port	Available

Table No :5.5.5 GSM Module Main Features




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5.6 ULTRASONIC SENSOR:

An ultrasonic sensor is a device that detects an object and measures the distance to it. It measures the distance by emitting ultrasound and receiving the wave that the object reflects.

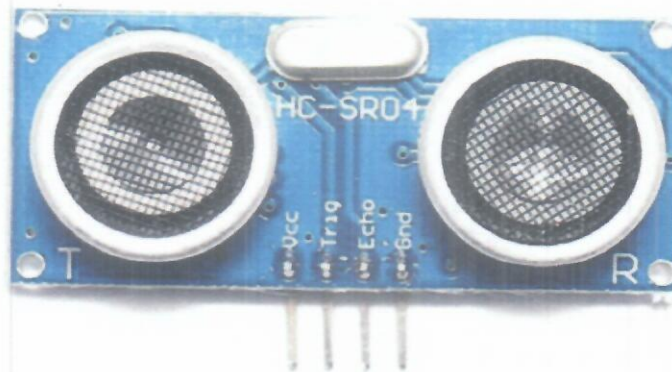


Fig No : 5.6 HC-SR04 Sensor

Working of an HC-SR04 Sensor:

Ultrasonic sound vibrates at a frequency above the range of human hearing. Transducers are the microphones used to receive and send ultrasonic sound. HC-SR04 and like other ultrasonic sensor module use a single transducer to send a pulse and to receive the echo. The sensor determines the distance to the target by measuring time lapse between sending and receiving of the ultrasonic pulses.

$$\text{Distance } L = \frac{1}{2} \times T \times C$$



where L is the distance, T is the time between the emission and reception, and C is the sonic speed.

$$\text{distance} = \frac{\text{speed of sound} \times \text{time taken}}{2}$$

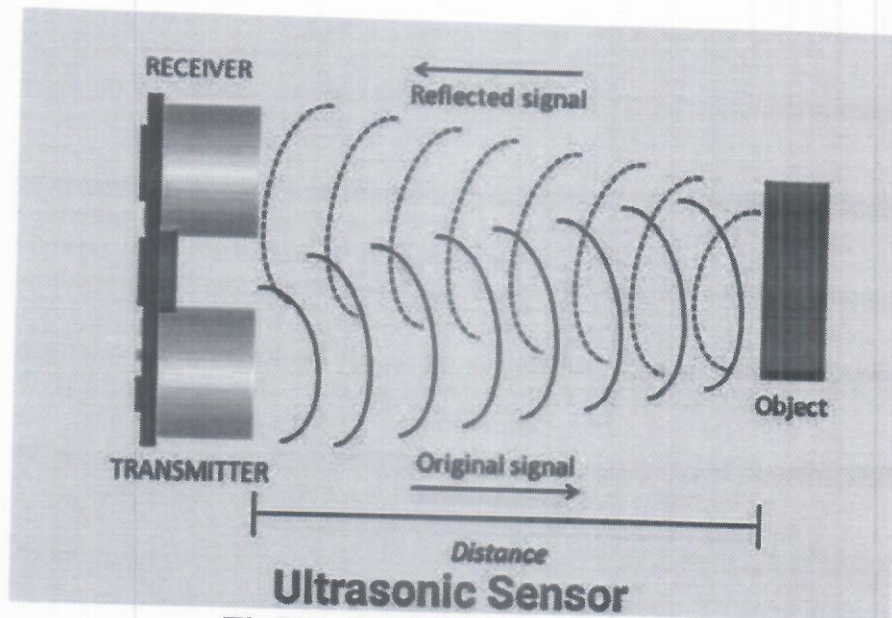


Fig No : 5.6.1 Ultrasonic Sensor Signals

An optical sensor has a transmitter and receiver, whereas an ultrasonic / level sensor uses a single ultrasonic element for both emission and reception. In a reflective model ultrasonic / level sensor, a single oscillator emits and receives ultrasonic waves alternately. This enables miniaturisation of the sensor head.



5.7 INTER-INTEGRATED CIRCUIT(I2C):

The Inter-Integrated Circuit (I²C) Protocol is a protocol intended to allow multiple "peripheral" digital integrated circuits ("chips") to communicate with one or more "controller" chips.

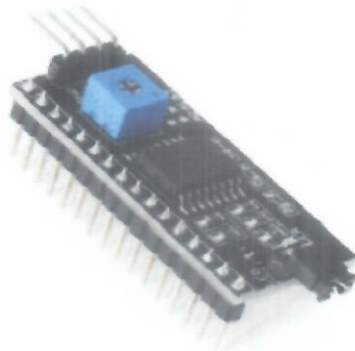


Fig No : 5.7 I2C BUS

The I2C 16×2 Arduino LCD Screen is using an I2C communication interface. It is able to display 16×2 characters on 2 lines, white characters on blue background.

Connection:

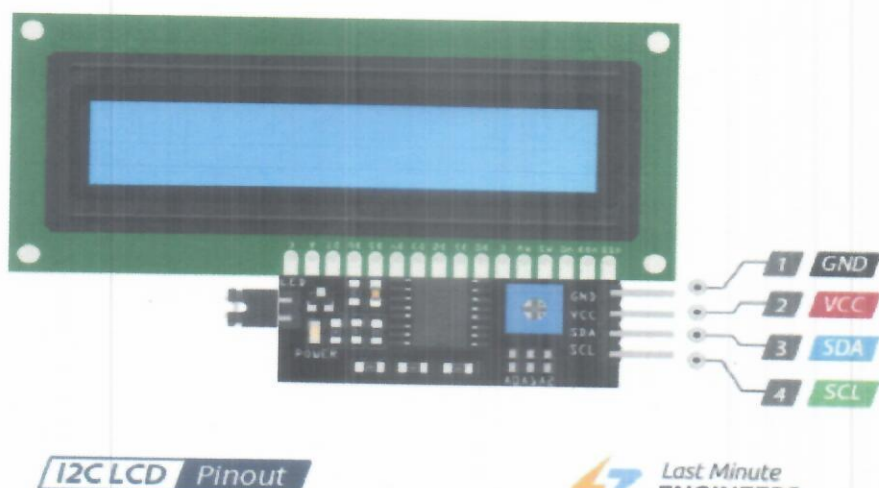


Fig No : 5.7.1 I2C Bus Connection

5.8 HUMIDITY SENSOR:

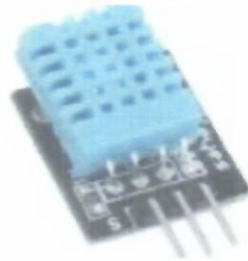


Fig No : 5.8 DHT11 Sensor

The DHT11 is a basic, ultra low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air, and spits out a digital signal on the data pin (no analog input pins needed). It's fairly simple to use, but requires careful timing to grab data.

Connection:

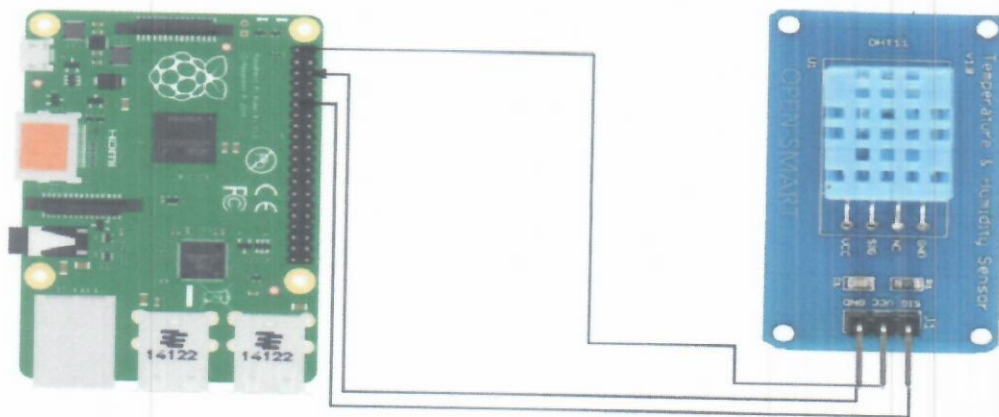


Fig No : 5.8.1 DHT11 Connection



5.9 ANDROID OVERVIEW

Android is an open-source platform founded in October 2003 by Andy Rubin and backed by Google, along with major hardware and software developers (such as Intel, HTC, ARM, Motorola and Samsung) that form the Open Handset Alliance. In October 2008, the software suite included on the phone consists of integration with Google's proprietary applications, such as Maps, Calendar, and Gmail, and a full HTML web browser.

Android supports the execution of native applications and third-party apps which are available via Google Play, which launched in October 2008 as Android Market. Nowadays android became the bestselling Smartphone. We now proceed to presenting an overview of the Android Platform and the components of an Android application.

As Shown in Figure 5.12, the Android platform is composed of 4 layers: Applications at the top, an Application Framework layer that provides services to applications, e.g., controlling activities or providing

- data access
- Library/VM layer at the bottom
- Linux kernel.




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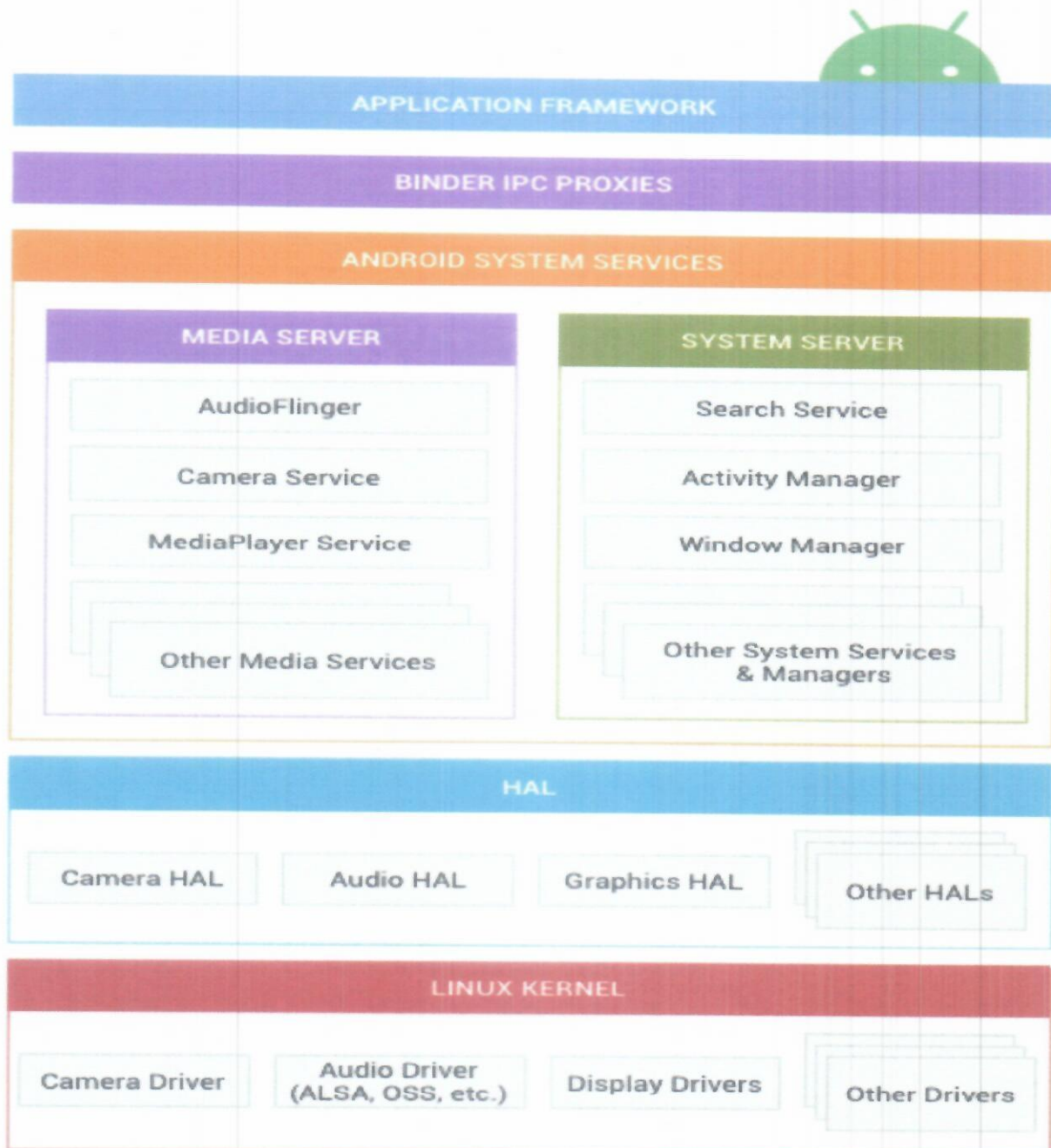


Fig No:5.9.1 Android Architecture



Android application interface:

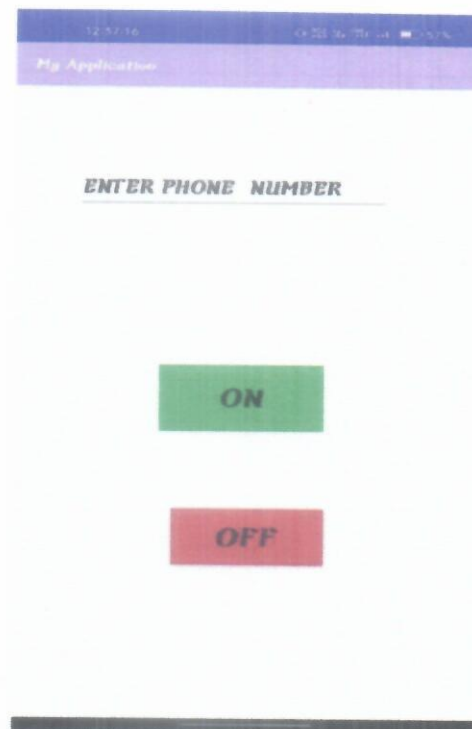


Fig No:5.9.1 Android Application Interface




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Android application interface:

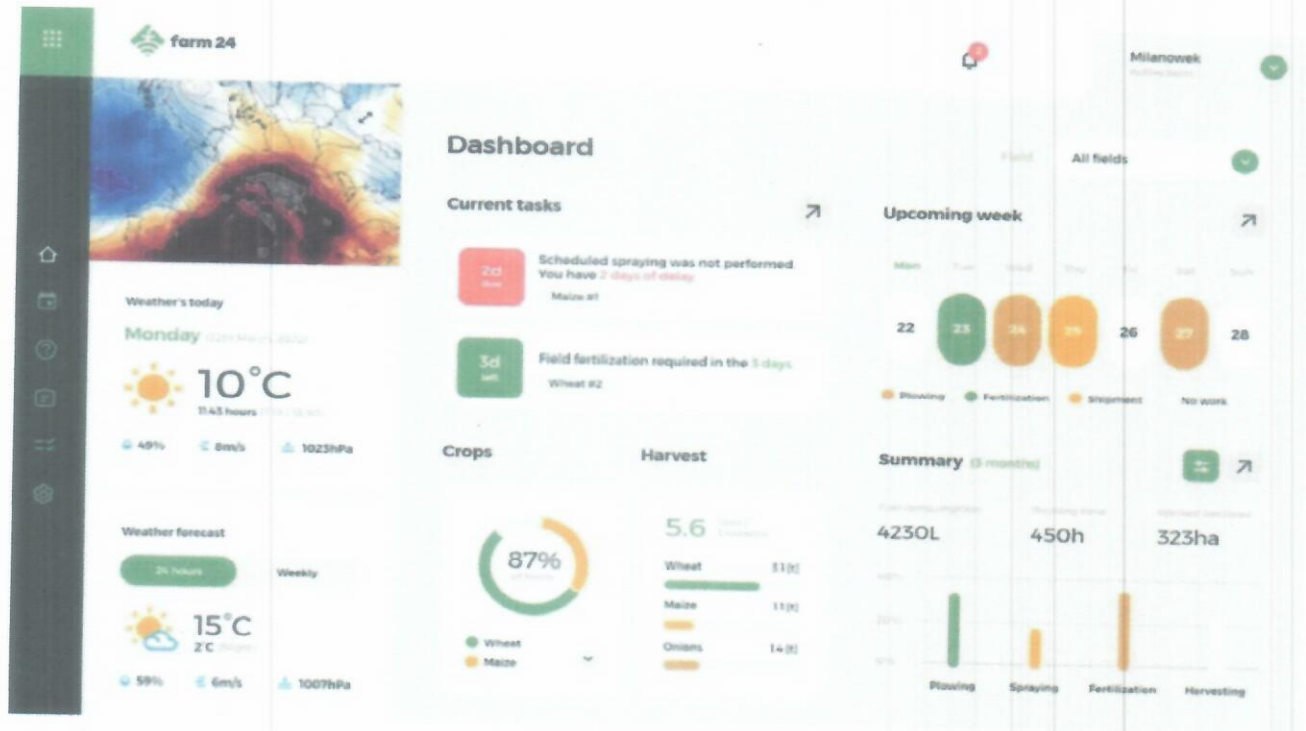


Fig No:5.9.2 Web Application



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GSM Module and mobile output:

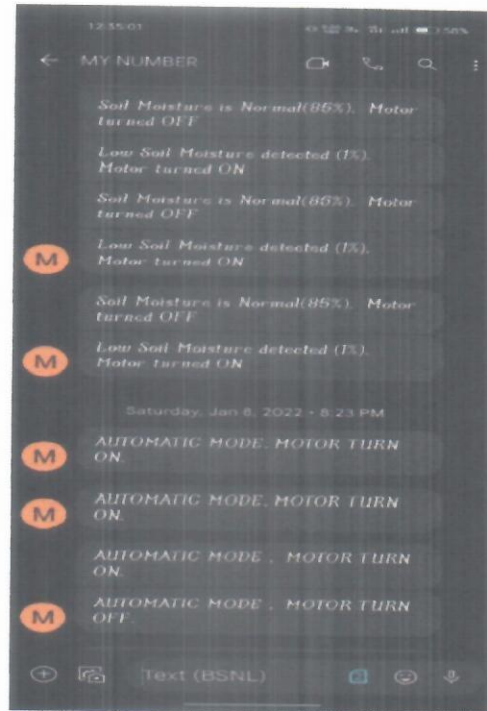


Fig No: 5.9.3 Messages

The motor on/off condition from GSM module to users mobile . If more than one agriculture fields are there, if the water is flowing in the first field ,after enough water is supplied the value of the motor gets off and automatically the value of the motor gets on in the second field and vise versa . the run time of the motor in field 1 and field 2 and if the motor gets off then the message is sent to the farmers mobile also.



CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

It can be seen that the combination of hardware and software provides a irrigation controller that can be implemented at relatively low cost and which is extremely user friendly. From the point of view of working at remote place the developed microcontroller based irrigation system can work constantly for indefinite time period, even in certain abnormal circumstances. If the plants get water at the proper time then it helps to increase the production from 25 to 30 %. This system can be used to irrigate very large areas as it only needs to divide the whole land into number of sectors and single microcontroller can control the whole process. It saves human energy, time, cost, etc.

6.2 FUTURE SCOPE

The proposed system consist of less hardware as compared to the previous model hence it is compact as compared to the previous system. It is more cost efficient, this claim is made on the fact that the proposed system does not need the heavy and expensive hardware for implementation. This type of automated irrigation system consumes 40-50% less water as compared to the traditional system Ideal growth condition is been provided when small amount of water is been applied over large amount of time. This smart irrigation system extends watering time for plants, and provides ideal growth condition. It saves time and timer delay as per the environmental condition can be added for automatic watering. This smart irrigation system can be adjusted and modified according to the changing environment. It is simple to operate it starts by designing the map of your garden and marking the location of planting.



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- [2] Mohamed Rawidean Mohd Kassim, Ibrahim Mat and Ahmad Nizar Harun, "Wireless Sensor Network in Precision agriculture application", international conference on *computer Information and telecommunication systems (CITS)* published in IEEE Xplore, July 2014.
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- [5] P. Rajalakshmi and S. Devi Mahalakshmi, "IOT Based Crop-Field Monitoring and Irrigation Automation", 10th International conference on *Intelligent systems and control (ISCO)* 7–8 Jan 2016 published in IEEE Xplore, Nov 2016.



APPENDIX

MAIN PROGRAM:

```
#include<SoftwareSerial.h>

SoftwareSerial Serial1(2,4);

#include <Wire.h>

#include <LiquidCrystal_I2C.h>

// Set the LCD address to 0x27 for a 16 chars and 2 line display
LiquidCrystal_I2C lcd(0x27, 16, 2);//SDA,SCL(A4,A5)

int led=13;

int led1Pin=9;//SOUND

int flag=0;

String str="";//rajandharan

#define motor 3

#define sensor 12

String textMessage ;

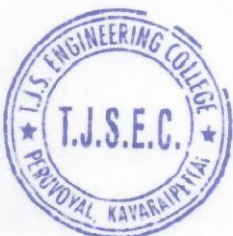
void setup()

{

    lcd.begin();//when using I2C to change ,lcd.begin();

    Serial1.begin(9600);

    Serial.begin(9600);
```




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```

pinMode(led1Pin, OUTPUT); //SOUND

pinMode(led, OUTPUT);

pinMode(motor, OUTPUT);

pinMode(sensor, INPUT_PULLUP);

String textMessage ;

Serial1.print ( "AT+CMGF=1\r" );

delay ( 100 );

// Set module to send SMS data to serial out upon receipt

Serial1.print ("AT+CNMI=2,2,0,0,0\r" );

delay ( 100 );

lcd.print("....");

lcd.setCursor(0,1);

lcd.print("...");

delay(2000);

lcd.clear();

lcd.print(".....");

lcd.setCursor(0,1);

lcd.print("....");

delay(2000);

lcd.clear();

lcd.setCursor(4,1);

```




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```

    lcd.clear();

    lcd.print(".....");

    lcd.setCursor(0,1);

    lcd.print("Welcomes You");

    delay(2000);

    gsmInit();

    lcd.clear();

    lcd.print("System Ready");
}

void loop()
{
    if ( Serial1 . available()> 0)
    {
        textMessage = Serial1 . readString();

        Serial. print ( textMessage );

        delay ( 10);

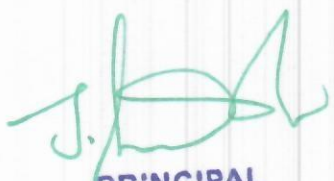
    }

    if ( textMessage . indexOf ( "ON" )>=0)
    {
        lcd.setCursor(0,0);

        lcd.print("SMS MODE      ");
    }

```




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```

lcd.setCursor(0,1);

lcd.print("Motor ON  ");

digitalWrite(motor, LOW);

if(digitalRead(sensor)==1 && flag==0)
{
    delay(1000);

    if(digitalRead(sensor)==1)
    {
        digitalWrite(led, LOW);

        sendSMS("Low Soil Moisture detected (1%). Motor turned ON");

        lcd.setCursor(0,1);

        lcd.print("Motor ON  ");

        digitalWrite(motor, LOW);

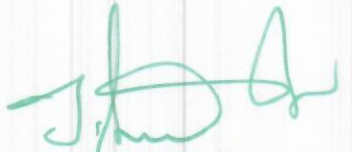
        delay(2000);

        flag=1;
    }
}

if(digitalRead(sensor)==0 && flag==1)
{
    delay(1000);

```




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```

if(digitalRead(sensor)==0)
{
    digitalWrite(led, HIGH);
    sendSMS("Soil Moisture is Normal(85%). Motor turned OFF");
    digitalWrite(motor, HIGH);
    lcd.setCursor(0,0);
    lcd.print("Automatic Mode ");
    lcd.setCursor(0,1);
    lcd.print("Motor OFF");
    delay(2000);
    flag=0;
}
}

if(digitalRead(sensor)==1 && flag==0)
{
    delay(1000);
    if(digitalRead(sensor)==1)
    {
        digitalWrite(led, LOW);
        sendSMS("Low Soil Moisture detected (1%). Motor turned ON");
        lcd.setCursor(0,0);
    }
}

```



[Handwritten Signature]

```

        lcd.print("Automatic Mode ");

        lcd.setCursor(0,1);

        lcd.print("Motor ON ");

        digitalWrite(motor, LOW);

        delay(2000);

        flag=1;

    }

}

if ( textMessage . indexOf ("OFF")>= 0 )

{

    digitalWrite(motor, HIGH);

    delay(2000);

    lcd.setCursor(0,0);

    lcd.print("SMS MODE ");

    lcd.setCursor(0,1);

    lcd.print("Motor OFF ");

}

}

void sendSMS(String msg)

{

    lcd.clear();

```




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```

    lcd.print("Sending SMS");

    Serial1.println("AT+CMGF=1");

    delay(500);

    Serial1.print("AT+CMGS=");

    Serial1.print("");

    Serial1.print("+916379919452"); // number

    Serial1.print("");

    Serial1.println();

    delay(500);

    Serial1.println(msg);

    delay(500);

    Serial1.write(26);

    delay(1000);

    lcd.clear();

    lcd.print("SMS Sent");

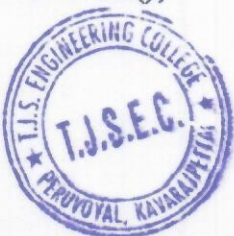
    delay(1000);

    lcd.begin();//when using I2C ,lcd.begin();
}

void gsmInit()
{

    lcd.clear();

```




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```

lcd.print("Finding Module..");

boolean at_flag=1;

while(at_flag)
{
    Serial1.println("AT");

    while(Serial1.available()>0)
    {
        if(Serial1.find("OK"))
            at_flag=0;
    }

    delay(1000);
}

Serial1.println("ATE0");

lcd.clear();

lcd.print("Finding Network..");

boolean net_flag=1;

while(net_flag)
{
    Serial1.println("AT+CPIN?");

    while(Serial1.available()>0)
    {

```



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```

    if(Serial1.find("READY"))

    net_flag=0;

    break;

}

delay(1000);

}

Serial1.println("AT+CNMI=2,2,0,0,0");

delay(1000);

Serial1.println("AT+CMGF=1");

delay(1000);

Serial1.println("AT+CSMP=17,167,0,0");

lcd.clear();

Serial1.flush();

}

```




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Chennai - 600 002. Ph: 90426 86793, 99412 77975, 044 4213 1795
Email: : khaleelrehman793@gmail.com, oceanrahman793@gmail.com Web: scienceprojectschennai.com

No.

Date: 12-03-22

M/s

No.	Particulars	Qty	Amount
(1)	Arduino Uno	4	3600/-
(2)	Lcd 16 ^x 2	1	150/-
(3)	Gsm 900A	1	1200/-
(4)	Node Mcu C ESP12D	1	600/-
(5)	Soil Monitor sensor	4	360/-
(6)	Relay 5V 5pin (channel)	1	200/-
(7)	Digital 3 phase meter		2800/-
	PAID!		
			/
		Total	8,910/-

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Email: : khaleelrehman793@gmail.com, oceanrahman793@gmail.com Web: scienceprojectschennai.com

No.

Date: 14/03/22

W/s

No.	Particulars	Qty	Amount
1.	Raspberry Pi	1	3000/-
2.	12C	1	120/-
3.	ultrasonic sensor	4	240/-
4.	PCB Board	3	130/-
5.	Load Lamp	1	500/-
6.	Solar controller	2	1020/-
7.	Solar panel 24V	1	3000/-
8.	wire	-	200/-
		Total	8210/-

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TAMILNADU STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

GOVERNMENT OF TAMILNADU



CERTIFICATE

This is to certify that **Mr./Ms. B. Karthick**, **T.J.S. Engineering College**, **Gummidipoondi- 601 206** has successfully completed the project titled “**Ai based automatic irrigation system with mobile application**” in the Sector **ELECTRICAL AND ELECTRONICS ENGINEERING** under **STUDENT PROJECT SCHEME** sponsored by the Council during the academic year 2022-2023.

Chennai-600 025

27.10.2023

EEE-2047/2023



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T.J.S. ENGINEERING COLLEGE

Peruvoyal, Kavarai, Tamil Nadu

Gummidipoondi- 601 206

July 2022 to June 2023

DR. R. SRINIVASAN
Member Secretary



TAMILNADU STATE COUNCIL FOR SCIENCE AND TECHNOLOGY

GOVERNMENT OF TAMILNADU



CERTIFICATE

This is to certify that **Mr./Ms. V. Kumaresan**, T.J.S. Engineering College, Gummidipoondi- 601 206 has successfully completed the project titled “Ai based automatic irrigation system with mobile application” in the Sector **ELECTRICAL AND ELECTRONICS ENGINEERING** under **STUDENT PROJECT SCHEME** sponsored by the Council during the academic year 2022-2023.

Chennai-600 025

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Thiruvallur Dist - 601 206.

DR. R. SRINIVASAN

Member Secretary